

Usage Manual

Beam & Stress — MechanicsOne. Eighteen screens, what each one is for, what it asks you for, what it gives back, and one worked task on every screen.

Finding your way around

Every screen is reachable from the list down the left-hand side. There are eighteen of them and they are always in the same order, so a screen you used yesterday is where you left it. Nothing is nested; nothing is behind a menu.

All eighteen are laid out the same way: **what you change** on the left, **what it does** in the middle, **what it comes to** on the right. Drag anything on the left and the picture and the numbers move in the same frame. On a narrow window the three fold into two and then one, and the pairing that survives folding is always slider-next-to-picture — because that pairing is the point.

Every screen opens on a worked set of numbers. Nothing is blank and nothing has to be filled in before the screen does something, so you can always start by dragging what is already there and watching what moves. Each screen's table below lists the value it opens on.

Conventions

- **Units.** Everything is held internally in SI and converted only where it is shown. Changing a unit picker moves the number, not just the label.
- **A dash for a unit means the quantity has none.** It is a ratio — a length over a length, a stress over a stress — so it is the same number whatever units you measured the two parts in. Slenderness, Poisson's ratio, the spring index, a factor of safety and joint efficiency are all of this kind. Strain is too, which is why it is usually quoted in millionths.
- **Downward is positive** for beam deflection, because that is the direction loads act and the direction you would sketch. A load pushing down is entered as a negative intensity.
- **A dash in a readout means there is no answer**, not that the answer is zero. It appears when an input is missing or the question has no answer as posed.
- **Refusals are results.** A load outside the span or a hole larger than the plate it is cut from produces a sentence explaining why, not a number and not a crash.
- **Choices are never made for you.** Where two answers are both defensible, both are shown with the reason beside them. Every screen below has a *Where you have to choose* heading; if it says there is nothing to choose, there really is nothing.

How the screens connect

Most screens stand alone, but two of them build something the others reuse. This is the one piece of structure worth learning first, because it is what lets you carry a single problem across several screens instead of typing the same section in four times.

- **Beam Diagrams** builds the beam — its span, supports and every load on it, with the shear and moment that follow. One later screen uses it without asking you again: *Beam Deflection*. Change it once and it moves.
- **Section Builder** builds the cross-section itself — its outline, area, centroid, second moments and extreme fibres. Seven later screens use it without asking you again: *Beam Diagrams*, *Beam Deflection*, *Combined Loading*, *Columns*, *Bending Stress*, *Shear Stress*, *Plastic Bending*. Change it once and every one of them moves.

Each screen below says at the top which of these it needs, so you can also just start anywhere and follow the pointer back.

Screens

Stress at a Point

Rotate the element and watch every component move together.

Where it is. 1st of 18 in the sidebar. It stands alone: nothing has to be set up before it.

WHAT IT IS FOR

Drag the angle and watch the element rotate, the circle point travel, and all three components move together.

$$\sigma_x' = (\sigma_x + \sigma_y)/2 + (\sigma_x - \sigma_y)/2 \cos 2\theta + \tau_{xy} \sin 2\theta$$
$$\sigma_{1,2} = (\sigma_x + \sigma_y)/2 \pm \sqrt{((\sigma_x - \sigma_y)/2)^2 + \tau_{xy}^2}$$

The plane of the element is not always the plane that fails. Ordering all three principal values decides it.

$$\tau_{abs} = (\sigma_{max} - \sigma_{min})/2 \text{ over } \{\sigma_1, \sigma_2, \sigma_3 = 0\}$$

The same circle as for stress, provided the shear strain enters halved. That halving is the commonest error in the subject.

$$\epsilon_x' = (\epsilon_x + \epsilon_y)/2 + (\epsilon_x - \epsilon_y)/2 \cos 2\theta + \gamma_{xy}/2 \sin 2\theta$$

Three normal components in, three out, plus the split of the stored energy that the distortion criterion depends on.

$$\epsilon_x = (1/E)[\sigma_x - \nu(\sigma_y + \sigma_z)]$$
$$\gamma_{xy} = \tau_{xy}/G$$
$$u = u_{vol} + u_{dist}$$

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
σ_x	MPa	—	80.00 MPa
σ_y	MPa	—	−40.00 MPa
τ_{xy}	MPa	—	50.00 MPa
σ_z	MPa	—	0 MPa
θ	deg	−90.0 to 90.0 deg	—
E	GPa	1.00 to 600 GPa	200.0 GPa
ν	— (a ratio)	−0.990 to 0.490	0.3000

A dash in the last two columns is a control that belongs to a row you add yourself — a load, a part, a segment — or one whose limits follow another input, so there is no single value to print. The screen still shows both as soon as the row exists.

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
$\sigma_{x'}$	MPa	Normal stress on a face rotated away from the reference direction.
$\tau_{x'y'}$	MPa	Shear stress on a face rotated away from the reference direction.
$\epsilon_{x'}$	– (a ratio)	Direct strain along a direction rotated away from the reference axis.
$\gamma_{x'y'}$	– (a ratio)	Engineering shear strain referred to rotated axes.
σ_1	MPa	
σ_2	MPa	
σ_{avg}	MPa	
θ_p	deg	
ϵ_x	– (a ratio)	
ϵ_y	– (a ratio)	
ϵ_z	– (a ratio)	
γ_{xy}	– (a ratio)	
u	Pa	
u_{vol}	Pa	
u_{dist}	Pa	
e	– (a ratio)	Fractional change in volume.
$\tau_{max, in\ plane}$	MPa	Both are shown at all times because they are not always the same number, and the larger one is the one that fails.
$\tau_{max, absolute}$	MPa	
$\gamma_{max, in\ plane}$	– (a ratio)	The same factor of two, in strain.
$\gamma_{max, absolute}$	– (a ratio)	

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	σ_x	Pa	Normal stress on the face whose outward normal points along the first axis.
IN	σ_y	Pa	Normal stress on the face whose normal points along the second axis.
IN	τ_{xy}	Pa	Shear stress on the first face acting along the second axis.
IN	θ	rad	Angle through which an element or a cutting plane has been rotated. Rotation of the element, measured anticlockwise
IN	σ_1	Pa	Largest of the principal stresses.
IN	σ_2	Pa	Middle or second principal stress.
IN	σ_3	Pa	Smallest principal stress; zero for a state of plane stress.
IN	ϵ_x	– (a ratio)	Direct strain along the first axis.
IN	ϵ_y	– (a ratio)	Direct strain along the second axis.
IN	γ_{xy}	– (a ratio)	Engineering shear strain: the total change in the right angle between two axes. Engineering shear strain, the full angle change
IN	σ_z	Pa	Normal stress on the face whose normal points along the third axis.
IN	E	Pa	Stiffness in simple tension: stress divided by strain while the response is still straight.
IN	ν	– (a ratio)	How much the material narrows sideways per unit of stretch along the pull.
IN	G	Pa	Stiffness in shear: shear stress divided by the angle change it produces.
OUT	σ_{xp}	Pa	Normal stress on a face rotated away from the reference direction.
OUT	τ_{xyp}	Pa	Shear stress on a face rotated away from the reference direction.
OUT	θ_p	rad	Angle to the plane on which the normal stress is greatest and the shear vanishes.
OUT	$\tau_{inplane_max}$	Pa	Largest shear stress on any plane containing the third axis.
OUT	σ_{avg}	Pa	Average of the two in-plane normal stresses; the centre of the stress circle.
OUT	θ_s	rad	Angle to the plane carrying the greatest in-plane shear stress.
OUT	$circle_R$	Pa	Radius of the stress circle; the largest in-plane shear stress.
OUT	σ_{sorted}	Pa	The three principal stresses put in descending order.
OUT	τ_{abs}	Pa	Largest shear stress on any plane at all, including those out of the element plane.
OUT	$governing_plane$	– (a ratio)	Which plane carries the largest shear stress.
OUT	ϵ_{xp}	– (a ratio)	Direct strain along a direction rotated away from the reference axis.
OUT	γ_{xyp}	– (a ratio)	Engineering shear strain referred to rotated axes.
OUT	ϵ_1	– (a ratio)	Largest principal strain.
OUT	ϵ_2	– (a ratio)	Smallest principal strain.
OUT	$\gamma_{inplane_max}$	– (a ratio)	Largest shear strain among planes containing the third axis.
OUT	γ_{abs_max}	– (a ratio)	Largest shear strain on any plane at all.
OUT	ϵ_z	– (a ratio)	Direct strain along the third axis.
OUT	e_{vol}	– (a ratio)	Fractional change in volume.
OUT	u_{total}	J/m ³	Total stored work per unit volume under a three-dimensional state.
OUT	u_{vol}	J/m ³	Part of the stored work that goes into changing volume.
OUT	u_{dist}	J/m ³	Part of the stored work that goes into changing shape.

WHERE YOU HAVE TO CHOOSE

Which maximum shear is being reported?

largest in the plane of the element | largest over all planes in three dimensions

For two principal stresses of the same sign these differ by a factor of two, in the unsafe direction. Show both, always.

Stress at a Point, the shear panel: tau max in plane and tau max absolute are shown as a pair, always, with a decision card beneath saying which plane carries the larger.

Do the in-plane principal stresses share a sign?

same sign: the out-of-plane pairing governs and tau equals half the larger | opposite signs: the in-plane pairing governs

This is a decision the user must see stated, not a hidden conditional. Getting it wrong under-predicts.

Stress at a Point, the state-of-stress panel: the sigma x, sigma y and sigma z sliders. The decision card in the shear panel names the pairing that governs and changes as the signs of the principal stresses do; the tau max absolute readout beside it changes with it.

Which two-dimensional idealisation applies?

plane stress: the out-of-plane stress vanishes | plane strain: the out-of-plane strain vanishes

These are different physical assumptions with different out-of-plane consequences; they must never be mixed on one screen.

Stress at a Point, the state-of-stress panel: the sigma z slider, with the decision card headed which idealisation applies stating plane stress while sigma z is zero, plane strain while the out-of-plane strain is, and neither otherwise. The epsilon z readout in the strain panel is the number the card is reading.

A WORKED TASK

A point in a loaded plate is pulled hard one way, pushed the other, and sheared as well. You want the worst tension it sees and the plane that tension acts on.

- 1. Type the three in-plane components into the left column.
- 2. Read the principal pair and the angle straight off the right column; the circle in the middle redraws as you type.
- 3. Drag the angle slider to that angle and watch the shear readout fall to zero — that is the check.

YOU TYPE		
σ_x	120.0 MPa	
σ_y	-40.00 MPa	
τ_xy	50.00 MPa	
IT COMES BACK WITH		
σ_1	134.3 MPa	the larger principal stress
σ_2	-54.34 MPa	the smaller one
θ_p	16.00 deg	rotate the element by this much and the shear vanishes
τ_max	94.34 MPa	the largest shear in the plane

The two principal values straddle the pair you typed, and the maximum shear is exactly half their difference. Both are worth checking by hand once.

Beam Diagrams

Shear and bending moment, redrawn as the loads move.

Where it is. 2nd of 18 in the sidebar. It reads the cross-section itself — its outline, area, centroid, second moments and extreme fibres from the Section Builder screen, so set that up first if you have not.

WHAT IT IS FOR

Set the supports, list the loads, read the two diagrams. The flagship screen.

$$dV/(dx)=-w(x) \quad dM/(dx)=V(x) \quad V(x)=\sum_i c_i \langle x-a_i \rangle^{n_i}$$

One redundant reaction, one compatibility condition. Solve it two ways and the answers must match.

$$M_{\text{(fix, udl)}}=wL^2/12 \quad M_{\text{mid}}=wL^2/24 \quad M_{\text{settle}}=6EI\delta/L^2$$

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
L	m	0.500 to 20.0 m	6.000 m
x	m	0 to 6.00 m	2.000 m
a, the left support	m	0 to 6.00 m	0 m
b, the right support	m	0 to 6.00 m	—
at	m	0 to 6.00 m	—
to	m	0 to 6.00 m	—

A dash in the last two columns is a control that belongs to a row you add yourself — a load, a part, a segment — or one whose limits follow another input, so there is no single value to print. The screen still shows both as soon as the row exists.

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
b – a, between the supports	m	
R_a	kN	Reaction at the left support.
R_b	kN	Reaction at the right support.
M_fix	kN·m	
V(x)	kN	
M(x)	kN·m	
M_max	kN·m	
at x	m	
V_max	kN	
released at the prop	kN	
released at the fixing	kN	
V just before	kN	The station sits on a discontinuity, so the shear there is two numbers and neither is an average of them.
V just after	kN	
M just before	kN·m	
M just after	kN·m	

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	span_L	m	Length of the beam from end to end, tails included. The same number as the distance between the supports only when the supports are at the two ends.
IN	support_type	– (a ratio)	Which support arrangement the beam has. simply supported, cantilever, or overhanging with one or two tails
IN	support_a	m	Where the left-hand support sits, measured from the left end of the beam. Where the left-hand support sits, measured from the left end of the beam. Zero for a beam supported at its ends.
IN	support_b	m	Where the right-hand support sits, measured from the left end of the beam. Where the right-hand support sits. Equal to span_L for a beam supported at its ends; anything less leaves a tail.
IN	loads	– (a ratio)	List of the loads acting on the beam, each with its type, magnitude and position. List of loads: point force, applied couple, uniform strip, or linearly varying strip
IN	a_pos	m	Position where a load starts. Station where a load starts
IN	b_pos	m	Position where a distributed load ends. Station where a distributed load ends
IN	P	N	Force applied along the member axis, or the single load in a standard case.
IN	M_couple	N·m	Couple applied directly to the beam at a point.
IN	w_int	N/m	Load carried per unit length of beam.
IN	x_station	m	Position along the beam at which a result is wanted.
IN	E	Pa	Stiffness in simple tension: stress divided by strain while the response is still straight.
IN	I_x	m ⁴	Second moment of area about the horizontal centroidal axis; how the material is spread vertically.
IN	δ_settle	m	Prescribed downward movement of a support. Prescribed support settlement, if any
IN	redundant_choice	– (a ratio)	Which restraint is released to make the structure determinate. Which restraint is released to make the beam determinate
OUT	R_a	N	Reaction at the left support.
OUT	R_b	N	Reaction at the right support.
OUT	V_x	N	Internal shear force at the chosen position along the beam.
OUT	M_x	N·m	Internal bending moment at the cut.
OUT	V_max	N	Largest internal shear force anywhere along the beam.
OUT	M_max	N·m	Largest internal bending moment anywhere along the beam.
OUT	x_at_Mmax	m	Position along the beam where the bending moment is greatest.
OUT	M_fix_a	N·m	Moment the left support must supply to keep the end from rotating.
OUT	M_fix_b	N·m	Moment the right support must supply to keep the end from rotating.
OUT	M_settle	N·m	Moment a built-in end carries because the two ends finished at different levels, before any load is applied.

WHERE YOU HAVE TO CHOOSE

Which sign convention should the diagrams use?

shear positive when it rotates the element clockwise | the opposite convention

Two of the four primary texts draw these upside down from each other. Offer the switch; never choose silently.

Beam Diagrams, the conventions panel: the switch reading shear positive clockwise. The shear diagram mirrors about the axis when it is thrown.

Which restraint is released?

the prop reaction | the fixing moment

Both are valid and must agree; letting the user switch makes the check a teaching device.

Beam Diagrams, the conventions panel: a two-segment control reading the prop / the built-in end, shown whenever the support chosen above it is indeterminate. The reactions are recomputed from the other release and must agree.

A WORKED TASK

A beam sits on two supports and carries a load spread evenly along the whole span. You want the largest bending moment and where it happens.

1. Leave the support picker on the simply supported case.
2. Set the span, then add one uniform load covering the whole beam.
3. The peak is marked on the moment diagram and printed beside it.

YOU TYPE

span	6.000 m	
w	-12.00 kN/m	downward, so the sign is negative

IT COMES BACK WITH

V_max	-36.00 kN	largest shear, at a support
M_max	-54.00 kN·m	largest bending moment
x_at_M	3.000 m	and where along the span it happens

Change the span alone and the moment follows the square of it: double the span and the moment goes up four times.

Section Builder

Assemble a section and read its properties as you go.

Where it is. 3rd of 18 in the sidebar. It stands alone, and what you build here is what the later screens work with.

WHAT IT IS FOR

Build a section from a palette of primitives, add or subtract each, and read off every property the rest of the app needs.

$$A=\sum A_i \quad \bar{y}=\sum A_i y_i/(\sum A_i) \quad I=\bar{I}+Ad^2 \quad I_p=I_x+I_y \quad r=\sqrt{I/A} \quad S=I/c$$

Rotate the axes until the product term vanishes. Structurally the same mathematics as the stress circle, deliberately written twice.

$$I_{(1,2)}=(I_x+I_y)/2\pm\sqrt{((I_x-I_y)/2)^2+I_{xy}^2} \quad \tan 2\theta_p=(-2I_{xy})/(I_x-I_y)$$

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
fibre y	mm	−500 to 500 mm	0 mm
axis θ	deg	−90.0 to 90.0 deg	—
place x	mm	−500 to 500 mm	—
place y	mm	−500 to 500 mm	—
rotate	deg	−90.0 to 90.0 deg	—

A dash in the last two columns is a control that belongs to a row you add yourself — a load, a part, a segment — or one whose limits follow another input, so there is no single value to print. The screen still shows both as soon as the row exists.

THE TEN SHAPES YOU CAN PLACE

Each part is one of these. The sliders you get change with the shape, because a circle does not have a width and a depth. Every part also takes a placement, a rotation, and a sign saying whether it adds material or removes it.

SHAPE	WHAT IT ASKS FOR
rectangle	width b, depth h
circle	radius r
hollow circle	outer radius, inner radius
right triangle	base b, height h
semicircle	radius r
quarter circle	radius r
ellipse	semi-axis a, semi-axis b
trapezoid	top width a, bottom width b, height h
thin ring	mean radius r, wall t
circular sector	radius r, half-angle

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
A	mm ²	Cross-sectional area actually carrying the load.
$\bar{x}$	mm	
$\bar{y}$	mm	Distance from a chosen reference line to the centroid of the whole section.
I _x	mm ⁴	
I _y	mm ⁴	
I _{xy}	mm ⁴	Product of inertia; zero whenever either reference axis is an axis of symmetry.
I _p	mm ⁴	
r _x	mm	
r _y	mm	
I ₁	mm ⁴	
I ₂	mm ⁴	
θ _p	deg	
I about θ	mm ⁴	
Q	mm ³	First moment about the neutral axis of the area lying beyond the level being examined.
t at the level	mm	
S at the top fibre	mm ³	They differ whenever the section is unsymmetric about its horizontal axis, and the smaller one governs.
S at the bottom fibre	mm ³	

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	parts	– (a ratio)	Ordered list of primitive shapes that together make up the section. Ordered list of primitives, each with shape, parameters, placement and a sign
IN	shape	– (a ratio)	Which primitive from the palette this part is. One of the ten primitives the palette places: rectangle, circle, hollow circle, right triangle, semicircle, quarter circle, ellipse, trapezoid, thin ring, circular sector. Every one is described by lengths alone except the sector, which takes a radius and a half-angle. The circular segment and the parabolic spandrel are named in the standard tables and are not offered.
IN	x_place	m	Horizontal position of a part relative to the section origin.
IN	y_place	m	Vertical position of a part relative to the section origin.
IN	rot	rad	Rotation of a part about its own centroid. Rotation of the primitive about its own centroid
IN	part_sign	– (a ratio)	Whether this part adds material or cuts a hole. Plus adds material, minus removes it
IN	y_fibre	m	Distance from the neutral axis to the fibre whose stress is wanted. Distance from the centroid to the fibre being examined
IN	I_x	m ⁴	Second moment of area about the horizontal centroidal axis; how the material is spread vertically.
IN	I_y	m ⁴	Second moment of area about the vertical centroidal axis.
IN	I_xy	m ⁴	Product of inertia; zero whenever either reference axis is an axis of symmetry. Zero whenever either axis is one of symmetry
IN	θ	rad	Angle through which an element or a cutting plane has been rotated. Rotation applied to the reference axes
OUT	A	m ²	Cross-sectional area actually carrying the load.
OUT	y_bar	m	Distance from a chosen reference line to the centroid of the whole section.
OUT	I_p	m ⁴	Polar second moment about the centroid; the sum of the two in-plane values.
OUT	r_gyr	m	Radius of gyration; the distance at which all the area could sit and give the same second moment.
OUT	S_mod	m ³	Section modulus; second moment divided by the distance to the extreme fibre.
OUT	Q_first	m ³	First moment about the neutral axis of the area lying beyond the level being examined.
OUT	I_1	m ⁴	Larger of the two principal second moments.
OUT	I_2	m ⁴	Smaller of the two principal second moments.
OUT	θ_p_I	rad	Angle to the axis about which the second moment is greatest.
OUT	I_x_rot	m ⁴	Second moment about an axis rotated away from the reference axis.

WHERE YOU HAVE TO CHOOSE

Does this part add or remove material?

add | remove (hole)

A hole entered as a negative dimension instead of a negative sign corrupts every downstream module.

Section Builder, each part carries its own two-segment control reading Add / Remove. Every section property -- area, centroid, both second moments and the product -- changes with it, and a hole is never a negative dimension.

A WORKED TASK

You need the properties of a T built from two rectangles: a flange lying across the top and a web hanging below it.

1. Add a rectangle, set its width and depth to the flange, and place it above the origin.
2. Add a second rectangle for the web and leave it on the origin.
3. Every property updates as you drag; the cross marks the centroid.

YOU TYPE		
b_f	200.0 mm	flange width
t_f	20.00 mm	flange thickness
b_w	20.00 mm	web thickness
d_w	200.0 mm	web depth
y_f	110.0 mm	where the flange centre sits
IT COMES BACK WITH		
A	8000 mm ²	area actually carrying the load
y_bar	55.00 mm	centroid, measured from where you placed the web
I_x	3.767e+07 mm ⁴	second moment about the centroidal axis
r_x	68.62 mm	radius of gyration

The centroid does not sit half way up: the flange drags it towards the top, and every stress on the screens that use this section is measured from there.

Beam Deflection

The third diagram for the same beam, with the peak located.

Where it is. 4th of 18 in the sidebar. It reads the beam — its span, supports and every load on it, with the shear and moment that follow and the cross-section itself — its outline, area, centroid, second moments and extreme fibres from the Beam Diagrams screen and the Section Builder screen, so set that up first if you have not.

WHAT IT IS FOR

Integrate the moment twice with singularity terms and the boundary conditions fix the constants.

EI d²v/(dx²)=M(x) ∫ ∫ (x-a)ⁿ dx=(∫ (x-a)ⁿ⁺¹)/(n+1)

Roughly twenty closed forms, each an independent check on the integrator and a source of printed fixtures.

cantilever, uniform strip: δ=wL⁴/(8EI) simple span, central load: δ=PL³/(48EI)

Linear elasticity means the cases simply add. Cheap, and a free cross-check on the integrator.

v(x)=Σᵢ vᵢ(x) θ(x)=Σᵢ θᵢ(x)

The area under the curvature diagram is the change in slope; its first moment is the deviation.

θ_BA=∫_A^B (M/(EI)) dx t_BA=∫_A^B M/(EI) x̄ dx

Differentiate the stored energy with respect to the load whose deflection is wanted.

δᵢ=∂ U/(∂ Pᵢ)=∫ (M/(EI)) ∂ M/(∂ Pᵢ) dx δ=Σ N n L/(AE)

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
E	GPa	1.00 to 600 GPa	200.0 GPa
station x	m	0 to 6.00 m	3.000 m
tangent drawn at	m	0 to 6.00 m	0 m
N, real force	kN	-500 to 500 kN	—
n, unit force	— (a ratio)	-3.00 to 3.00	—
L	m	0.100 to 20.0 m	—
A·E	MN	1.00 to 2000 MN	—
δ, how far it drops	mm	0 to 50.0 mm	0 mm
P	kN	-200 to 200 kN	20.00 kN
w	kN/m	-100 to 100 kN/m	4.000 kN/m
M	kN·m	-200 to 200 kN·m	15.00 kN·m
a	m	0 to 6.00 m	2.000 m

A dash in the last two columns is a control that belongs to a row you add yourself — a load, a part, a segment — or one whose limits follow another input, so there is no single value to print. The screen still shows both as soon as the row exists.

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
span	m	
loads	– (a ratio)	
I _x	mm ⁴	
EI	N·m ²	
v(x)	mm	Sideways movement of the beam axis at a given position.
θ(x)	deg	Angle the deflected beam axis makes with its original direction.
v _{max}	mm	
at x	m	Position along the beam where the deflection is greatest.
Δθ	deg	Change in slope accumulated between two positions.
t, deviation	mm	Vertical distance from a point on the beam to the tangent drawn at another point.
v(x), moment-area	mm	
v(x), conjugate beam	mm	
v(x), virtual work	mm	
U, bending	J	Total work stored in the whole member.
the three routes spread by	%	
δ, joint	mm	Movement of a joint found by summing the stretch of every member.
M _{settle}	kN·m	
from the closed form	mm	
from the integrator	mm	
v _{max} , closed form	mm	
θ _{max} , closed form	deg	

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	span_L	m	Length of the beam from end to end, tails included. The same number as the distance between the supports only when the supports are at the two ends.
IN	loads	– (a ratio)	List of the loads acting on the beam, each with its type, magnitude and position.
IN	E	Pa	Stiffness in simple tension: stress divided by strain while the response is still straight.
IN	I_x	m ⁴	Second moment of area about the horizontal centroidal axis; how the material is spread vertically.
IN	x_station	m	Position along the beam at which a result is wanted.
IN	support_type	– (a ratio)	Which support arrangement the beam has.
IN	P	N	Force applied along the member axis, or the single load in a standard case.
IN	w_int	N/m	Load carried per unit length of beam.
IN	M_couple	N·m	Couple applied directly to the beam at a point.
IN	a_pos	m	Position where a load starts.
IN	case_id	– (a ratio)	Which standard configuration from the closed-form table is wanted.
IN	cases	– (a ratio)	List of standard load cases to be added together. List of standard cases to be summed
IN	x_ref	m	Position used as the reference when taking a tangential deviation. Reference station
IN	A	m ²	Cross-sectional area actually carrying the load.
IN	G	Pa	Stiffness in shear: shear stress divided by the angle change it produces.
IN	members	– (a ratio)	List of straight axial members with their forces, lengths and stiffnesses.
IN	include_shear	– (a ratio)	Whether shear energy is counted alongside bending energy.
OUT	v_defl	m	Sideways movement of the beam axis at a given position.
OUT	θ_slope	rad	Angle the deflected beam axis makes with its original direction.
OUT	v_max	m	Largest deflection anywhere along the beam.
OUT	x_at_vmax	m	Position along the beam where the deflection is greatest.
OUT	θ_max	rad	Largest slope anywhere along the beam.
OUT	θ_change	rad	Change in slope accumulated between two positions.
OUT	t_deviation	m	Vertical distance from a point on the beam to the tangent drawn at another point.
OUT	U_total	J	Total work stored in the whole member.
OUT	δ_truss	m	Movement of a joint found by summing the stretch of every member.

WHERE YOU HAVE TO CHOOSE

Nothing on this screen has two defensible answers, so there is nothing to choose.

A WORKED TASK

The same beam again — simply supported, evenly loaded — and on the same section you built two screens ago. Now you want how far it sags.

1. Nothing to retype: the beam comes from Beam Diagrams and the section from Section Builder.
2. Give the screen the material, and the rigidity follows from the section's own second moment.
3. The deflected shape is drawn under the moment diagram and the peak is located for you.

YOU TYPE

span	6.000 m	
w	–12.00 kN/m	
E	200.0 GPa	steel

IT COMES BACK WITH

v_max	–26.88 mm	largest sag, at mid-span
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Deflection goes as the fourth power of the span. That is why a beam that is strong enough can still be too bendy.

Combined Loading

Five actions at once, and the point on the section that suffers most.

Where it is. 5th of 18 in the sidebar. It reads the cross-section itself — its outline, area, centroid, second moments and extreme fibres from the Section Builder screen, so set that up first if you have not.

WHAT IT IS FOR

Axial force, two bending moments, shear and torque acting at once. The closing screen, and the whole course in one picture.

$$\sigma=N/A\pm M_z y/I_z\pm M_y z/I_y \quad \tau=VQ/(It)+Tp/J$$

(added as vectors, not magnitudes)

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
y	mm	−500 to 500 mm	0 mm

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
z, where that puts you	mm	
the worst point is at y	mm	
and at z	mm	
σ	MPa	Normal stress on the face whose outward normal points along the first axis.
τ	MPa	Shear stress on the first face acting along the second axis.
σ_1	MPa	
σ_2	MPa	
σ_{eq} where you are looking	MPa	The scan runs whatever you choose, so these two are never the same computation.
σ_{eq} at the scanned worst	MPa	
at the extreme fibre	MPa	
at the neutral axis	MPa	

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	N_axial	N	Internal force acting along the member axis at the cut being examined.
IN	V_y	N	Shear force component in the first transverse direction.
IN	M_y_mom	N·m	Bending moment component about the first bending axis.
IN	M_z_mom	N·m	Bending moment component about the second bending axis.
IN	T_torque	N·m	Twisting moment about the member axis.
IN	A	m ²	Cross-sectional area actually carrying the load.
IN	I_y	m ⁴	Second moment of area about the vertical centroidal axis.
IN	I_z	m ⁴	Second moment of area about the axis used as the bending axis in unsymmetric work.
IN	J_polar	m ⁴	Polar second moment of a round section; sets both twist and shear stress.
IN	Q_first	m ³	First moment about the neutral axis of the area lying beyond the level being examined.
IN	t_at_y	m	Width of the section at the level where the shear stress is being taken.
IN	y_fibre	m	Distance from the neutral axis to the fibre whose stress is wanted.
IN	z_fibre	m	Distance from the second bending axis to the point whose stress is wanted.
IN	ρ_rad	m	Radius of the particular point being examined.
OUT	σ_x	Pa	Normal stress on the face whose outward normal points along the first axis.
OUT	τ_xy	Pa	Shear stress on the first face acting along the second axis.
OUT	critical_point	– (a ratio)	Point on the section that proves to be worst once every action is combined.
OUT	σ_1	Pa	Largest of the principal stresses.
OUT	σ_eq_distortion	Pa	Equivalent stress under the distortion-energy criterion.

WHERE YOU HAVE TO CHOOSE

Which point on the section is being checked?

extreme fibre: bending peaks, transverse shear vanishes | neutral axis: bending vanishes, transverse shear peaks | the scanned worst point, which is usually neither

The final skill the course teaches. Show the user's chosen point and the scanned worst point side by side.

Combined Loading, the panel headed the point you are checking: the fibre position slider. The panel headed your point against the scan puts the scanned worst point beside it with a decision card saying which governs, and the two obvious candidates panel holds the extreme fibre and the neutral axis together.

A WORKED TASK

A bracket carries a pull, a bend and a twist at once. You want the single number to compare against a yield stress.

1. Set the five actions on the left.
2. The screen finds the point on the section that suffers most, rather than assuming it is the top fibre.
3. The equivalent stress is printed with the criterion named.

YOU TYPE		
σ	90.00 MPa	direct stress at that point
τ	45.00 MPa	shear at the same point
IT COMES BACK WITH		
σ_1	108.6 MPa	
σ_2	-18.64 MPa	
σ_eq	119.1 MPa	equivalent stress, distortion energy

Compare that last number, not the direct stress, against the yield stress of the material.

Failure Criteria

Five envelopes on one plane, and where your state sits.

Where it is. 6th of 18 in the sidebar. It stands alone: nothing has to be set up before it.

WHAT IT IS FOR

Five envelopes on one plane, with the current state as a draggable point. The static figure in every textbook, made live.

$$\sigma_v = \sqrt{\left(\frac{1}{2}\right) [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]}$$
$$\sigma_{\max} - \sigma_{\min} = \sigma_Y$$

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
v	– (a ratio)	–0.990 to 0.490	0.3000

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
envelope ratio	– (a ratio)	How far the state has gone towards the envelope built from separate tensile and compressive strengths.
governing criterion		
n	– (a ratio)	How many times the applied action could be increased before failure.

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	σ_1	Pa	Largest of the principal stresses.
IN	σ_2	Pa	Middle or second principal stress.
IN	σ_3	Pa	Smallest principal stress; zero for a state of plane stress.
IN	σ_Y	Pa	Stress at which the material stops behaving elastically.
IN	v	– (a ratio)	How much the material narrows sideways per unit of stretch along the pull.
IN	σ_ut	Pa	Greatest stress the material carries in tension before it breaks. Ultimate strength in tension, for the brittle criterion
IN	σ_uc	Pa	Greatest stress the material carries in compression before it breaks. Ultimate strength in compression
OUT	σ_eq_max_normal	Pa	Single stress that the largest-normal-stress criterion treats as equivalent to the actual state.
OUT	σ_eq_max_shear	Pa	Equivalent stress under the largest-shear criterion.
OUT	σ_eq_distortion	Pa	Equivalent stress under the distortion-energy criterion.
OUT	σ_eq_max_strain	Pa	Equivalent stress under the largest-normal-strain criterion.
OUT	σ_eq_total_energy	Pa	Equivalent stress under the total-stored-energy criterion.
OUT	brittle_ratio	– (a ratio)	How far the state has gone towards the envelope built from separate tensile and compressive strengths.
OUT	n_safety	– (a ratio)	How many times the applied action could be increased before failure.

WHERE YOU HAVE TO CHOOSE

Which criterion governs the check?

maximum normal stress | maximum shear stress | maximum distortion energy | maximum normal strain | total strain energy

They disagree by up to 15 percent for ductile metals and far more for brittle materials. Report all five side by side.

Failure Criteria, the panel headed equivalent stress, five ways: all five criteria are listed together against the same state, with the governing one and its factor of safety named beneath and all five envelopes drawn on one plane.

A WORKED TASK

A point is in tension one way and compression the other. Five criteria disagree about how close to yield it is, and you want to see by how much.

- 1. Type the two principal stresses.
- 2. All five envelopes are drawn on one plane with your state marked as a point.
- 3. The equivalent stress for each criterion is listed beside the plot.

YOU TYPE		
σ_1	150.0 MPa	
σ_2	-60.00 MPa	
ν	0.3000	
σ_Y	250.0 MPa	yield stress
IT COMES BACK WITH		
normal	150.0 MPa	maximum normal stress
shear	210.0 MPa	maximum shear stress
distortion	187.3 MPa	distortion energy
n	1.334	factor of safety on the distortion-energy value

The criteria differ by a noticeable margin here. Which one you are entitled to use is a property of the material, not of the loading — and the screen will not choose for you.

Columns

Slenderness, eccentricity, and the curve that governs.

Where it is. 7th of 18 in the sidebar. It reads the cross-section itself — its outline, area, centroid, second moments and extreme fibres from the Section Builder screen, so set that up first if you have not.

WHAT IT IS FOR

A slender column fails by going sideways long before the material yields. Which way it goes is a decision the user must see.

$$P_{cr}=\pi^2EI/(KL)^2 \quad \sigma_{cr}=\pi^2E/(KL/r)^2 \quad (KL/r)_c=\sqrt{(2\pi^2E/\sigma_Y)}$$

The one relation in the app that cannot be inverted in closed form. Finding the allowable load needs a bracketed root search.

$$\sigma_{max}=(P/A)[1+ec/r^2 \sec ((L/(2r))\sqrt{P/(EA)})] \quad \delta=e[\sec ((L/(2r))\sqrt{P/(EA)})-1]$$

Short columns yield before they buckle. The parabola is built to meet the ideal curve tangentially.

$$\sigma_{cr}=\sigma_Y[1-\sigma_Y (KL/r)^2/(4\pi^2E)] \quad P=\sigma_c A/(1+a (L_e/r)^2) \quad P_t=\pi^2E_t I/(KL)^2$$

Axial load amplifies whatever bending is already there, without bound as the critical load is approached.

$$M_{max}=M_0 \sec ((L/2)\sqrt{P/(EI)}) \quad \text{amplification}=1/(1-P/P_{cr})$$

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
L	m	0.200 to 12.0 m	3.000 m
E	GPa	1.00 to 600 GPa	200.0 GPa
σ_Y	MPa	50.0 to 900 MPa	250.0 MPa
K, strong axis	– (a ratio)	0.500 to 2.50	1.000
K, weak axis	– (a ratio)	0.500 to 2.50	1.000
e c / r²	– (a ratio)	0 to 2.00	0.4000
σ_allow	MPa	50.0 to 900 MPa	150.0 MPa
Rankine 1/a	– (a ratio)	1000 to 30000	–
E tangent	GPa	5.00 to 220 GPa	120.0 GPa
P, axial	kN	0 to 500 kN	50.00 kN
M _θ , end couple	kN·m	0 to 50.0 kN·m	4.000 kN·m
a _θ , initial bow	mm	0 to 50.0 mm	3.000 mm

A dash in the last two columns is a control that belongs to a row you add yourself — a load, a part, a segment — or one whose limits follow another input, so there is no single value to print. The screen still shows both as soon as the row exists.

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
K for that condition	– (a ratio)	Factor converting real length into the length of an equivalent pin-ended column.
λ	– (a ratio)	Effective length divided by radius of gyration; how slender the column is.
λ_c	– (a ratio)	Slenderness at which yielding and buckling become equally likely.
σ_{cr} , elastic	MPa	
σ_{cr} , parabolic	MPa	
σ_{cr} , governing	MPa	
P failure	N	
P _{allow}	N	Largest load an eccentrically loaded column may carry at the allowable stress.
P Rankine–Gordon	kN	Column capacity from the empirical form that blends crushing into buckling.
P tangent modulus	kN	Buckling load found using the local slope of the stress-strain curve instead of the initial one.
P Euler, weak axis	kN	
amplification	– (a ratio)	How much axial load magnifies bending already present.
u, secant argument	rad	The angle whose secant amplifies the moment. It reaches a quarter turn exactly at the buckling load, which is why the moment grows without bound there.
M max, exact	kN·m	Largest internal bending moment anywhere along the beam.
M max, by amplification	kN·m	
δ at mid-span	mm	Sideways movement at the middle of a member.
extra movement	mm	How much further the mid-point moves than it already sat, once the axial load acts on the bow that was there to begin with.
M from the bow	kN·m	The bending the axial load creates by pushing along a member that is not quite straight. Zero for a perfectly straight one, which is why the ideal column is an idealisation.
slenderness, strong axis	– (a ratio)	Effective length over radius of gyration, for each axis with its own K.
slenderness, weak axis	– (a ratio)	
P _{cr} , strong axis	N	Both are ideal elastic loads, one per axis, and the smaller of the two is the one to compare. Whether the column can reach it is the next panel.
P _{cr} , weak axis	N	

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	E	Pa	Stiffness in simple tension: stress divided by strain while the response is still straight.
IN	I _x	m ⁴	Second moment of area about the horizontal centroidal axis; how the material is spread vertically.
IN	I _y	m ⁴	Second moment of area about the vertical centroidal axis.
IN	A	m ²	Cross-sectional area actually carrying the load.
IN	L	m	Length of the member along its axis.
IN	K _x	– (a ratio)	Effective-length factor for buckling about the stronger axis.
IN	K _y	– (a ratio)	Effective-length factor for buckling about the weaker axis.
IN	end_condition	– (a ratio)	How the two ends of the column are held. pinned-pinned, fixed-fixed, fixed-pinned or fixed-free
IN	σ _Y	Pa	Stress at which the material stops behaving elastically.
IN	P	N	Force applied along the member axis, or the single load in a standard case.
IN	e _{ecc}	m	Distance from the centroid to the line of action of the load.
IN	c _{extreme}	m	Distance from the neutral axis to the farthest fibre.
IN	r _{gyr}	m	Radius of gyration; the distance at which all the area could sit and give the same second moment.
IN	σ _{allow}	Pa	Highest normal stress permitted in service.
IN	K _{eff}	– (a ratio)	Factor converting real length into the length of an equivalent pin-ended column.
IN	E _{tangent}	Pa	Local slope of the stress-strain curve at a point past the straight part.
IN	slenderness	– (a ratio)	Effective length divided by radius of gyration; how slender the column is.
IN	σ _{crush}	Pa	Compressive strength used by the empirical short-column form. Crushing strength used by the empirical hyperbolic form
IN	rankine_a	– (a ratio)	Empirical constant fixing where the hyperbolic column form bends over. Empirical constant of the hyperbolic form
IN	P _{cr}	N	Load at which an ideal straight column first goes sideways.
IN	M _{couple}	N·m	Couple applied directly to the beam at a point.
IN	a _{initial}	m	Initial bow already present at mid-length before any load is applied. Initial bow at mid-span
OUT	σ _{cr}	Pa	Average stress in a column at the moment it buckles.
OUT	slenderness _c	– (a ratio)	Slenderness at which yielding and buckling become equally likely.
OUT	governing_axis	– (a ratio)	Which axis actually controls the buckling capacity.
OUT	σ _{max}	Pa	Largest normal stress present.
OUT	δ _{mid}	m	Sideways movement at the middle of a member.
OUT	P _{allow}	N	Largest load an eccentrically loaded column may carry at the allowable stress.
OUT	σ _{cr_parabolic}	Pa	Column strength from the parabolic form used below the critical slenderness.
OUT	P _{hyperbolic}	N	Column capacity from the empirical form that blends crushing into buckling.
OUT	P _{tangent}	N	Buckling load found using the local slope of the stress-strain curve instead of the initial one.
OUT	amp_factor	– (a ratio)	How much axial load magnifies bending already present.
OUT	u _{secant}	rad	The angle whose secant amplifies the moment. It reaches a quarter turn exactly at the buckling load, which is why the moment grows without bound there.
OUT	M _{max}	N·m	Largest internal bending moment anywhere along the beam.
OUT	δ _{extra}	m	How much further the mid-point moves than it already sat, once the axial load acts on the bow that was there to begin with.
OUT	M _{bow}	N·m	The bending the axial load creates by pushing along a member that is not quite straight. Zero for a perfectly straight one, which is why the ideal column is an idealisation.

WHERE YOU HAVE TO CHOOSE

About which axis does the column buckle?

strong axis | weak axis | compare both, because the bracing differs between them

Taking the smaller second moment is right only when both axes share the same effective length. With intermediate bracing on one axis the comparison must be made explicitly, and getting it wrong over-predicts capacity.

Columns, the end-conditions panel: the K, strong axis and K, weak axis sliders. The panel headed which axis buckles shows both slendernesses and both critical loads as pairs, with a decision card naming the governing axis and saying so explicitly when the bracing is what decides it.

A WORKED TASK

A pin-ended strut is long enough to buckle before it squashes. You want the load that buckles it.

- 1. Give the screen the section — or inherit one.
- 2. Set the length and pick the end conditions.
- 3. The governing axis is named, because a section that is stiff one way buckles the other.

YOU TYPE		
E	200.0 GPa	
I	4.000e+06 mm ⁴	second moment about the weak axis
L	6.000 m	
A	2400 mm ²	
r	40.82 mm	radius of gyration, weak axis
σ_Y	250.0 MPa	yield stress of the steel
IT COMES BACK WITH		
P_cr	219.3 kN	buckling load, pinned both ends
λ	147.0	slenderness ratio
σ_cr	91.36 MPa	stress at that load
λ_c	125.7	above this slenderness the elastic curve is the honest one

Check the last two lines against each other before you trust the first: this strut is slender enough that it really does buckle elastically. Shorten it below the critical slenderness and the elastic load becomes an overestimate, which is why the screen draws both curves and marks where they cross. Change the end condition to fixed-fixed and the buckling load goes up four times, without a single dimension changing.

Torsion

Open against closed, and how much a slit costs.

Where it is. 8th of 18 in the sidebar. It stands alone: nothing has to be set up before it.

WHAT IT IS FOR

Shear grows linearly from the axis. Everything else on this screen follows from that one fact.

$$\tau = T\rho/J \quad J_{\text{solid}} = \pi c^4/2 \quad J_{\text{tube}} = (\pi/2)(c_o^4 - c_i^4) \quad \phi = TL/(JG)$$

Rated power and speed give the torque; the allowable shear gives the diameter.

$$P_w = T\omega = 2\pi f T \quad d = (16T/(\pi\tau_{\text{allow}}))^{1/3}$$

Shafts in series share the torque; shafts in parallel share the angle.

$$\phi = \sum_i T_i L_i / (J_i G_i) \quad \text{series: } T \text{ common} \quad \text{parallel: } \phi \text{ common}$$

Plane sections no longer stay plane. Two tabulated coefficients carry all the geometry.

$$\tau_{\text{max}} = T/(\alpha a b^2) \quad \phi = TL/(\beta a b^3 G) \quad \alpha, \beta = f(a/b)$$

Shear flow is constant around a closed loop, whatever the wall does.

$$\tau = T/(2tA_m) \quad q = \tau t = T/(2A_m) \quad \phi = (TL/(4A_m^2 G)) \oint ds/t$$

One slit along the length removes almost all the torsional stiffness. The comparison is the lesson.

$$J \approx (1/3) \sum_i b_i t_i^3 \quad \tau_{\text{max}} = 3T/(\sum b_i t_i^2)$$

Reduce a combined action to an equivalent single action, size for both, and take the larger diameter.

$$M_e = (1/2)(M + \sqrt{M^2 + T^2}) \quad T_e = \sqrt{M^2 + T^2} \quad d = (16T_e/(\pi\tau_{\text{allow}}))^{1/3}$$

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
read the stress at,	– (a ratio)	0 to 1.00	1.000
T	kN·m	–50.0 to 50.0 kN·m	3.000 kN·m
L	m	0.100 to 10.0 m	1.500 m
G	GPa	5.00 to 120 GPa	80.00 GPa
P	kW	1.00 to 2000 kW	100.0 kW
n, rev/s	– (a ratio)	0.500 to 300	50.00
τ_allow	MPa	5.00 to 400 MPa	60.00 MPa
M, bending	kN·m	–50.0 to 50.0 kN·m	2.000 kN·m
σ_allow	MPa	5.00 to 600 MPa	100.0 MPa
G·J of the second, as a fraction	– (a ratio)	0.0500 to 5.00	0.5000
torque applied at, as a fraction of span	– (a ratio)	0.0500 to 0.950	0.4000
T	kN·m	–50.0 to 50.0 kN·m	3.000 kN·m
L	m	0.100 to 10.0 m	1.500 m
d	mm	2.00 to 1000 mm	–
G	GPa	5.00 to 120 GPa	80.00 GPa

A dash in the last two columns is a control that belongs to a row you add yourself — a load, a part, a segment — or one whose limits follow another input, so there is no single value to print. The screen still shows both as soon as the row exists.

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
τ_{max}	MPa	Shear stress produced by twisting.
ϕ	deg	Angle one end of a shaft rotates relative to the other.
τ at ρ	MPa	Shear stress produced by twisting.
ρ	mm	
J	m ⁴	Polar second moment of a round section; sets both twist and shear stress.
GJ	N·m ²	
T _{allow}	N·m	Largest torque the shaft may carry at the allowable shear stress.
q	N/m	Shear force transferred per unit length along the member.
T	N·m	Twisting moment about the member axis.
d required	mm	Smallest diameter that satisfies the governing allowable stress.
P recovered	kW	
a/b	— (a ratio)	
α	— (a ratio)	Tabulated coefficient fixing the peak shear stress in a rectangular shaft.
β	— (a ratio)	Tabulated coefficient fixing the twist of a rectangular shaft.
M equivalent	N·m	Single bending moment that would be as severe as the actual bending and torque together.
T equivalent	N·m	Single torque that would be as severe as the actual bending and torque together.
d by normal stress	mm	
d by shear stress	mm	
ϕ total	rad	Angle one end of a shaft rotates relative to the other.
T first	N·m	Share of the torque taken by one shaft of a parallel pair.
T second	N·m	
sum	N·m	
T left	N·m	Torque supplied by a fixed end.
T right	N·m	
residual	N·m	
open / closed	— (a ratio)	
J, closed	m ⁴	Same wall, same radius, same material.
J, slit open	m ⁴	

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	T_torque	N·m	Twisting moment about the member axis.
IN	ρ_rad	m	Radius of the particular point being examined.
IN	c_out	m	Outer radius of a shaft.
IN	c_in	m	Inner radius of a hollow shaft.
IN	L	m	Length of the member along its axis.
IN	G	Pa	Stiffness in shear: shear stress divided by the angle change it produces.
IN	τ_allow	Pa	Highest shear stress permitted in service.
IN	power_w	W	Rate at which the shaft transmits work.
IN	freq_hz	Hz	Rotations per second.
IN	ω	rad/s	Rotation rate in radians per second.
IN	bore_ratio	– (a ratio)	Bore diameter divided by outer diameter.
IN	segments	– (a ratio)	List of prismatic pieces making up a stepped member. List of shaft segments, each with T, L, J and G
IN	a_long	m	Longer side of a rectangular shaft section. Longer side of the rectangle
IN	b_short	m	Shorter side of a rectangular shaft section.
IN	t_wall	m	Wall thickness of a tube or shell.
IN	A_median	m ²	Area enclosed by the line running along the middle of a tube wall.
IN	perimeter_m	m	Length once around the mid-line of a tube wall.
IN	strips	– (a ratio)	List of narrow rectangular walls making up an open thin-walled section. List of rectangular wall strips, each with length b and thickness t
IN	M_x	N·m	Internal bending moment at the cut.
IN	σ_allow	Pa	Highest normal stress permitted in service.
OUT	J_polar	m ⁴	Polar second moment of a round section; sets both twist and shear stress.
OUT	τ_torsion	Pa	Shear stress produced by twisting.
OUT	φ_twist	rad	Angle one end of a shaft rotates relative to the other.
OUT	Zp	m ³	Polar section modulus: polar second moment divided by the outer radius.
OUT	GJ	N·m ²	Torsional rigidity: the torque needed per unit twist per unit length.
OUT	T_allow	N·m	Largest torque the shaft may carry at the allowable shear stress.
OUT	d_req	m	Smallest diameter that satisfies the governing allowable stress.
OUT	T_split	N·m	Share of the torque taken by one shaft of a parallel pair.
OUT	T_reaction	N·m	Torque supplied by a fixed end.
OUT	α_rect	– (a ratio)	Tabulated coefficient fixing the peak shear stress in a rectangular shaft.
OUT	β_rect	– (a ratio)	Tabulated coefficient fixing the twist of a rectangular shaft.
OUT	J_eff	m ⁴	Effective torsion constant of a solid non-circular section.
OUT	q_flow	N/m	Shear force transferred per unit length along the member.
OUT	J_open	m ⁴	Effective torsion constant of an open thin-walled section.
OUT	stiffness_ratio	– (a ratio)	How many times stiffer a closed section is than the same section slit open.
OUT	M_equiv	N·m	Single bending moment that would be as severe as the actual bending and torque together.
OUT	T_equiv	N·m	Single torque that would be as severe as the actual bending and torque together.
OUT	governing_criterion	– (a ratio)	Which allowable stress sets the required size.

WHERE YOU HAVE TO CHOOSE

Which radius is the stress wanted at?

outer surface (maximum) | bore surface | an arbitrary radius

For a tube the bore stress is not zero, and reporting only the peak hides that.

Torsion, the shaft panel: the read the stress at slider, which runs from the bore to the outer surface. The tau at rho readout under it follows, and for a tube it does not reach zero at the bore.

Which allowable governs the size?

the normal-stress limit | the shear-stress limit

Reporting only the number hides which limit is binding, and a change of material can switch it.

Torsion, the panel headed sized for bending and torsion together: d by normal stress and d by shear stress on consecutive lines, the required diameter beneath them, and a decision card naming which limit is binding.

A WORKED TASK

A solid circular shaft carries a torque. You want the surface shear stress and how far one end turns relative to the other.

- 1. Leave the section on the solid circular case.
- 2. Set the diameter, the torque, the length and the shear modulus.
- 3. The stress varies linearly from zero at the axis to the printed value at the surface.

YOU TYPE		
r	30.00 mm	shaft radius
T	3.000 kN·m	
L	1.500 m	
G	80.00 GPa	
IT COMES BACK WITH		
J	1.272e+06 mm ⁴	polar second moment of the section
τ	70.74 MPa	shear stress at the surface
φ	2.533 deg	twist over the whole length

Slit that shaft along its length and it loses almost all of its torsional stiffness. The open-section case on this same screen shows by how much.

Bending Stress

The distribution across the section, and where the neutral axis went.

Where it is. 9th of 18 in the sidebar. It reads the cross-section itself — its outline, area, centroid, second moments and extreme fibres from the Section Builder screen, so set that up first if you have not.

WHAT IT IS FOR

Stress varies linearly from the neutral axis, which passes through the centroid because it has to.

$$\sigma = -My/I \quad \sigma_{\max} = Mc/I = M/S \quad 1/\rho = M/(EI)$$

When the loading plane misses a principal axis the neutral axis is not perpendicular to it.

$$\sigma = -M_z y/I_z + M_y z/I_y \quad (\text{principal axes}) \quad \tan \alpha = I_z/I_y \tan \theta$$

Trade one material for an equivalent width of the other, then use the ordinary flexure result.

$$n = E_2/E_1 \quad b' = n \quad b \quad \bar{y} = \sum \bar{A}_i \bar{y}_i / (\sum \bar{A}_i) \quad \sigma_1 = -My/I_t, \quad \sigma_2 = -nMy/I_t$$

Move the load off the centroid far enough and one face goes into tension. The boundary is the core.

$$\sigma = P/A \pm Pec/I \quad \text{no tension while } e \leq r^2/c \quad \text{rectangle: } b/6 \times h/6$$

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
M_x	kN·m	−500 to 500 kN·m	40.00 kN·m
M_y	kN·m	−500 to 500 kN·m	0 kN·m
fibre y	mm	−500 to 500 mm	0 mm
fibre z	mm	−500 to 500 mm	0 mm
N	kN	—	0 kN
e	mm	−500 to 500 mm	0 mm
E	GPa	1.00 to 600 GPa	200.0 GPa
σ_allow	MPa	1.00 to 1000 MPa	165.0 MPa

A dash in the last two columns is a control that belongs to a row you add yourself — a load, a part, a segment — or one whose limits follow another input, so there is no single value to print. The screen still shows both as soon as the row exists.

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
σ at the fibre	MPa	Normal stress produced by bending.
κ	1/m	Curvature of the bent axis: how sharply it bends per unit length.
S required	mm ³	
neutral axis α	deg	Angle the neutral axis makes with the reference axis under unsymmetric bending.
A	mm ²	
I _x	mm ⁴	
I _{xy}	mm ⁴	
n	– (a ratio)	
b transformed	mm	
I transformed	mm ⁴	
y _{bar}	mm	Where the neutral axis of a stacked two-material section falls, measured from the outer face of the first layer. Not at mid-depth, which is the whole difference from the side-by-side case.
σ , the other way round	MPa	The first material's stress, reached by transforming into the second instead.
σ at the top face	MPa	Equal and opposite only when the section is symmetric about its horizontal axis.
σ at the bottom face	MPa	
two-term form	MPa	
general form	MPa	
core, towards the top	mm	r^2/c at each face. For a rectangle these come out as the middle third; for a circle, a quarter of the diameter.
core, towards the bottom	mm	
σ in the first	MPa	Each at its own material's extreme fibre. They differ by the modular ratio, because the two materials are at the same strain and stress follows stiffness.
σ in the second	MPa	

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	M _x	N·m	Internal bending moment at the cut.
IN	y _{fibre}	m	Distance from the neutral axis to the fibre whose stress is wanted.
IN	I _x	m ⁴	Second moment of area about the horizontal centroidal axis; how the material is spread vertically.
IN	c _{extreme}	m	Distance from the neutral axis to the farthest fibre.
IN	σ _{allow}	Pa	Highest normal stress permitted in service.
IN	E	Pa	Stiffness in simple tension: stress divided by strain while the response is still straight.
IN	M _{y_mom}	N·m	Bending moment component about the first bending axis.
IN	M _{z_mom}	N·m	Bending moment component about the second bending axis.
IN	I _y	m ⁴	Second moment of area about the vertical centroidal axis.
IN	I _z	m ⁴	Second moment of area about the axis used as the bending axis in unsymmetric work.
IN	I _{yz}	m ⁴	Product of inertia in the coordinate pair used for unsymmetric bending.
IN	z _{fibre}	m	Distance from the second bending axis to the point whose stress is wanted.
IN	θ	rad	Angle through which an element or a cutting plane has been rotated.
IN	E1 _{mat}	Pa	Stiffness of the first material in a two-material assembly.
IN	E2 _{mat}	Pa	Stiffness of the second material.
IN	parts	– (a ratio)	Ordered list of primitive shapes that together make up the section.
IN	layout_composite	– (a ratio)	Which way the two materials sit: side by side across the width, both running the full depth, or one stacked on a face of the other. Side by side, or one layer stacked on a face of the other.
IN	d _{layer}	m	Depth of the layer bonded to a face of the first one, measured the same way the beam is deep. Only the stacked arrangement has a second depth; side by side both materials share one.
IN	P	N	Force applied along the member axis, or the single load in a standard case.
IN	A	m ²	Cross-sectional area actually carrying the load.
IN	e _{ecc}	m	Distance from the centroid to the line of action of the load.
IN	r _{gyr}	m	Radius of gyration; the distance at which all the area could sit and give the same second moment.
IN	b _{width}	m	Width of a rectangular section.
IN	h _{depth}	m	Overall depth of a section measured in the bending direction.
OUT	σ _{bend}	Pa	Normal stress produced by bending.
OUT	σ _{max}	Pa	Largest normal stress present.
OUT	S _{req}	m ³	Smallest section modulus that keeps the bending stress within the allowable value.
OUT	κ	1/m	Curvature of the bent axis: how sharply it bends per unit length.
OUT	σ _{bend_gen}	Pa	Bending stress from the general form that keeps the product-of-inertia term.
OUT	α _{na}	rad	Angle the neutral axis makes with the reference axis under unsymmetric bending.
OUT	n _{modular}	– (a ratio)	Ratio of the two stiffnesses, used to trade one material for the other.
OUT	b _{transformed}	m	Width the second material is redrawn as so the section behaves as one material.
OUT	I _{transformed}	m ⁴	Second moment of the section after one material has been traded for an equivalent width of the other.
OUT	σ _{1m}	Pa	Stress in the first material of a composite member.
OUT	σ _{2m}	Pa	Stress in the second material of a composite member.
OUT	y _{na_layered}	m	Where the neutral axis of a stacked two-material section falls, measured from the outer face of the first layer. Not at mid-depth, which is the whole difference from the side-by-side case.
OUT	I _{layered}	m ⁴	Second moment of a stacked two-material section, once the second layer has been redrawn at the first material's stiffness.
OUT	σ _{comb}	Pa	Normal stress from axial force and bending taken together.

OUT	e_kern	m	Largest eccentricity that leaves the whole section in compression.
OUT	kern_outline	m	Boundary of the region in which the load may act without causing tension.
OUT	no_tension	– (a ratio)	Whether the whole section stays in compression under the given eccentricity.

WHERE YOU HAVE TO CHOOSE

Are the reference axes principal?

yes: use the two-term form | no, the product term is non-zero: use the general form

Using the principal form when the product term is not zero puts the neutral axis in the wrong place entirely.

Bending Stress, the panel headed which form applies: a decision card naming the form, both forms evaluated side by side, the neutral-axis angle, and beneath them a second card saying whether the point being read can be the one that governs. What decides the form is the section's product of inertia, which the rotate slider on each part in the Section Builder moves off zero; the point itself is placed by the fibre y and fibre z sliders in the bending panel, and both coordinates are needed because the second term of the relation is proportional to z.

Which material is the section transformed into?

material 1 | material 2

Both are correct and must give the same physical answer; the intermediate numbers differ by the modular ratio.

Bending Stress, the flitched beam panel: a two-segment control naming which material the section is transformed into. The modular ratio inverts and the transformed second moment changes with it, while the stresses in the real materials do not.

How do the two materials sit -- side by side across the width, or one stacked on a face of the other?

side by side | stacked on a face

Side by side, both materials run the full depth, the section is symmetric and its neutral axis is at mid-depth however stiff the layers are. Stacked, they do not and it is not: the neutral axis moves towards the stiffer layer. The side-by-side closed form has room for only one depth, so a stacked section cannot be stated in it at all; handed the total depth, which is the only depth there is to give, it answers for a different beam -- seven times the second moment on the timber-and-strap case, with every intermediate number looking reasonable.

Bending Stress, the composite beam panel: a two-segment control at the top naming how the layers sit. Choosing the stacked arrangement reveals a second depth input and a neutral-axis readout, and the transformed second moment changes with it.

Is the eccentricity about one axis or two?

uniaxial | biaxial

Biaxial eccentricity needs both bending terms; a uniaxial screen quietly under-predicts the corner stress.

Bending Stress: the e slider sets the eccentricity about one axis and drives the two face-stress readouts and the core limits; the M_y slider adds the second bending term and moves the neutral-axis angle off the vertical. With M_y at zero the state is uniaxial.

A WORKED TASK

A beam section carries a bending moment. You want the largest stress in it.

1. Inherit the section, or set the second moment and the extreme fibre distance directly.
2. Type the moment.
3. The distribution is drawn across the depth with the neutral axis marked.

YOU TYPE

M	45.00 kN·m	
I	8.500e+07 mm ⁴	
c	150.0 mm	centroid to the extreme fibre

IT COMES BACK WITH

$\sigma_{\max}$	79.41 MPa	largest bending stress
σ_{half}	–39.71 MPa	half way out from the axis, on the other face

The second reading is half the size of the first, because bending stress is linear in the distance from the neutral axis. Its sign is opposite because it is the face on the other side of that axis: one is in tension, the other in compression.

Shear Stress

Shear through the depth, with the jump at every width step.

Where it is. 10th of 18 in the sidebar. It reads the cross-section itself — its outline, area, centroid, second moments and extreme fibres from the Section Builder screen, so set that up first if you have not.

WHAT IT IS FOR

Shear stress is zero at the free surfaces and peaks where the first moment does.

$\tau = VQ/(It)$ $\tau_{(max,rect)} = 3V/(2A)$ $\tau_{(max,circ)} = 4V/(3A)$

Built-up sections stay built up only if the fasteners carry the flow between the pieces.

$q = VQ/I$ $s = n_f F_{allow}/q$

Load a channel through its centroid and it twists. The point that avoids twist is somewhere outside the section.

$e_{channel} = b^2 h^2 t / (4I_z)$ $e_{(l, unequal)} = h I_1 / (I_1 + I_2)$ $e_{(l, offset)} = h^2 t (b_2^2 - b_1^2) / (4I_z)$

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
v	kN	−500 to 500 kN	50.00 kN
level y	mm	−500 to 500 mm	0 mm
capacity of one	kN	1.00 to 200 kN	12.00 kN
top flange	mm	10.0 to 400 mm	120.0 mm
bottom flange	mm	10.0 to 400 mm	75.00 mm
flange b ₁ , short side	mm from the web's centre line	0 to 400 mm from the web's centre line	60.00 mm from the web's centre line
flange b ₂ , long side	mm from the web's centre line	0 to 400 mm from the web's centre line	90.00 mm from the web's centre line
flange b	mm	10.0 to 400 mm	75.00 mm
web h	mm	20.0 to 800 mm	200.0 mm
wall t	mm	1.00 to 40.0 mm	8.000 mm
load acts at	mm from the centre	−200 to 200 mm from the centre	0 mm from the centre

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
τ	MPa	Shear stress produced by the shear force crossing the beam.
Q	mm ³	First moment about the neutral axis of the area lying beyond the level being examined.
t	mm	
τ_{max}	MPa	
at y	mm	
q	/m	Shear force transferred per unit length along the member.
spacing	mm	Longest fastener spacing that still carries the shear flow.
force in one fastener	kN	
e	mm	Distance from the section to the point through which load must pass to avoid twist.
$\int \tau t \, dy$	kN	
applied V	kN	
τ on the wide side	MPa	One level, one first moment, two widths. The ratio of the two stresses is exactly the ratio of the widths.
τ on the narrow side	MPa	

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	V	N	Internal shear force at the cut.
IN	Q_first	m ³	First moment about the neutral axis of the area lying beyond the level being examined.
IN	I_x	m ⁴	Second moment of area about the horizontal centroidal axis; how the material is spread vertically.
IN	t_at_y	m	Width of the section at the level where the shear stress is being taken. Width of the section at the level being examined
IN	A	m ²	Cross-sectional area actually carrying the load.
IN	y_fibre	m	Distance from the neutral axis to the fibre whose stress is wanted.
IN	F_fastener	N	Force a single fastener may carry.
IN	n_rows	– (a ratio)	Number of fastener rows sharing the shear flow.
IN	b_flange	m	Width of a flange measured out from the web.
IN	h_web	m	Depth of the web between flange centre lines.
IN	t_wall	m	Wall thickness of a tube or shell.
IN	I_z	m ⁴	Second moment of area about the axis used as the bending axis in unsymmetric work.
IN	I_1	m ⁴	Larger of the two principal second moments.
IN	I_2	m ⁴	Smaller of the two principal second moments.
IN	b_proj_short	m	The shorter of the two distances a flange reaches out sideways from the web's centre line, when the web does not sit in the middle of it.
IN	b_proj_long	m	The longer of those two distances, on the other side of the web.
OUT	τ_shear	Pa	Shear stress produced by the shear force crossing the beam.
OUT	τ_max	Pa	Largest shear stress present in the case at hand.
OUT	τ_pair	– (a ratio)	Which of the two shear stresses at a level where the section width steps is being read, and by what factor they differ -- the ratio of the two widths, since one first moment is divided by each of them in turn.
OUT	q_flow	N/m	Shear force transferred per unit length along the member.
OUT	s_spacing	m	Longest fastener spacing that still carries the shear flow.
OUT	F_per_fastener	N	Force each fastener actually carries at the chosen spacing.
OUT	e_shear	m	Distance from the section to the point through which load must pass to avoid twist.
OUT	e_shear_web_offset	m	Sideways distance from the web's centre line to the shear centre, for an I-section whose flanges match but whose web is off-centre in them.
OUT	twist_flag	– (a ratio)	Whether the load passes through the shear centre and therefore causes no twist.

WHERE YOU HAVE TO CHOOSE

Which side of a width discontinuity?

the wide side | the narrow side

The two differ by the ratio of the widths, which for a rolled beam is an order of magnitude.

Shear Stress, the at the level panel: where the width steps, the wide-side and narrow-side stresses appear as a pair, with a decision card giving the ratio between them; where it does not, the card says so and one value is shown.

How many fastener rows resist the flow?

one | two

A factor of two directly on the permitted spacing.

Shear Stress, the fasteners panel: a two-segment control reading one row / two rows. The permitted spacing readout doubles.

A WORKED TASK

The same beam carries a shear force as well. You want the largest shear stress through the depth of a rectangle.

1. Set the shear force and the rectangle.
2. The parabola is drawn through the depth; the peak is at the neutral axis, not at the surface.

YOU TYPE

V	60.00 kN	
A	30000 mm ²	area of the rectangle

IT COMES BACK WITH

$\tau_{\max}$	3.000 MPa	peak shear stress, at the neutral axis
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It is half again the average V/A . For a rectangle the factor is always three halves, whatever the size.

Strain Rosette

Three readings in, a full strain state out.

Where it is. 11th of 18 in the sidebar. It stands alone: nothing has to be set up before it.

WHAT IT IS FOR

Three gauge readings go in, the full plane strain state comes out. The laboratory workflow this replaces takes five error-prone minutes.

$$\epsilon_{\theta} = \epsilon_x \cos^2 \theta + \epsilon_y \sin^2 \theta + \gamma_{xy} \sin \theta \cos \theta \cos 45^\circ: \gamma_{xy} = 2\epsilon_b - \epsilon_a - \epsilon_c$$

WHAT YOU SET

Nothing to drag on this screen — it works from what the screens before it have already built.

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
determinant	– (a ratio)	Determinant of the three-gauge system; near zero means the angles do not fix a unique answer.
ϵ_x	– (a ratio)	Direct strain along the first axis.
ϵ_y	– (a ratio)	Direct strain along the second axis.
γ_{xy}	– (a ratio)	Engineering shear strain: the total change in the right angle between two axes.
ϵ_1	– (a ratio)	
ϵ_2	– (a ratio)	
θ_p	deg	
ϵ_a recovered	– (a ratio)	
ϵ_b recovered	– (a ratio)	
ϵ_c recovered	– (a ratio)	

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	eps_a	– (a ratio)	Reading from the first gauge of a rosette.
IN	eps_b	– (a ratio)	Reading from the second gauge.
IN	eps_c	– (a ratio)	Reading from the third gauge.
IN	ang_a	rad	Angle of the first gauge from the reference axis.
IN	ang_b	rad	Angle of the second gauge.
IN	ang_c	rad	Angle of the third gauge.
OUT	ϵ_x	– (a ratio)	Direct strain along the first axis.
OUT	ϵ_y	– (a ratio)	Direct strain along the second axis.
OUT	γ_{xy}	– (a ratio)	Engineering shear strain: the total change in the right angle between two axes.
OUT	rosette_det	– (a ratio)	Determinant of the three-gauge system; near zero means the angles do not fix a unique answer.

WHERE YOU HAVE TO CHOOSE

Nothing on this screen has two defensible answers, so there is nothing to choose.

A WORKED TASK

Three gauges are cemented to a surface at nought, forty-five and ninety degrees. You have three readings and want the full strain state.

1. Pick the rosette layout.
2. Type the three readings in microstrain.
3. The reduced state comes back with the principal strains and the circle drawn.

YOU TYPE		
a	520.0 ustrain	gauge at nought
b	360.0 ustrain	gauge at forty-five
c	-80.00 ustrain	gauge at ninety
θ_a	0 deg	
θ_b	45.00 deg	
θ_c	90.00 deg	

IT COMES BACK WITH		
eps_x	520.0 ustrain	direct strain along x
eps_y	-80.00 ustrain	direct strain along y
γ_{xy}	280.0 ustrain	shear strain

With this layout the first and third readings are the two direct strains outright; only the shear needs the middle gauge. Move the angles off the standard set and all three start mixing.

Reinforced Concrete

Cracked below the axis, and the quadratic that puts the axis there.

Where it is. 12th of 18 in the sidebar. It stands alone: nothing has to be set up before it.

WHAT IT IS FOR

Concrete is assumed cracked below the neutral axis, so the depth of the compression block solves a quadratic.

$$b \frac{h^2}{2} = n A_{st}(d-h') \quad I_{cr} = b \frac{h^3}{3} + n A_{st}(d-h')^2$$

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
b	mm	100 to 1000 mm	300.0 mm
d, to the steel	mm	100 to 1500 mm	500.0 mm
As	mm ²	100 to 10000 mm ²	1500 mm ²
n, modular ratio – (a ratio)		5.00 to 20.0	9.000
M	kN·m	1.00 to 1000 kN·m	120.0 kN·m

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
x	mm	Depth of the compression block in a cracked reinforced section.
I_cr	mm ⁴	Second moment of the transformed section once the concrete below the neutral axis is ignored.
z, lever arm	mm	
ρ	%	
x, the root the section keeps	mm	
the root the section rejects	mm	
σ concrete	MPa	Peak compressive stress in the concrete.
σ steel	MPa	Tensile stress in the reinforcement.
C, concrete	kN	
T, steel	kN	
C·z recovered	kN·m	
the two routes differ by	nm	
b x ² /2 – n As (d – x), at the first	m ³	Both vanish. That is what makes this a branch rather than an arithmetic step: the equation is satisfied by both numbers, so it cannot be the equation that rules one out.
at the second	m ³	

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	b_width	m	Width of a rectangular section.
IN	d_eff	m	Depth from the compression face down to the centre of the reinforcement. Depth from the top face to the steel
IN	A_steel	m ²	Total area of the reinforcing bars.
IN	n_modular	– (a ratio)	Ratio of the two stiffnesses, used to trade one material for the other.
IN	M_x	N·m	Internal bending moment at the cut.
OUT	h_na	m	Depth of the compression block in a cracked reinforced section.
OUT	I_cracked	m ⁴	Second moment of the transformed section once the concrete below the neutral axis is ignored.
OUT	σ_conc	Pa	Peak compressive stress in the concrete.
OUT	σ_steel	Pa	Tensile stress in the reinforcement.

WHERE YOU HAVE TO CHOOSE

Which root of the quadratic is physical?

the positive root | the negative root

Only one root lies inside the section; show the rejected root and why it is rejected.

Reinforced Concrete, the panel headed which root of the quadratic: both roots are shown on consecutive lines, the physical one marked and the other carrying the sentence that says it is a depth above the compression face.

A WORKED TASK

A rectangular concrete beam is reinforced with steel near the bottom and has cracked below the neutral axis. You want the stress in the concrete and in the steel.

- 1. Set the breadth, the effective depth and the steel area.
- 2. Set the modular ratio.
- 3. The neutral axis depth comes from a quadratic and is drawn on the section.

YOU TYPE		
b	300.0 mm	breadth
d	500.0 mm	centroid of steel to the top face
As	1500 mm ²	area of steel
n	9.000	steel modulus over concrete modulus
M	80.00 kN·m	

IT COMES BACK WITH		
kd	171.9 mm	neutral axis, measured down from the top face
I_cr	1.961e+09 mm ⁴	second moment of the cracked section
σ_c	7.010 MPa	top of the concrete, in compression
σ_s	120.5 MPa	in the steel, in tension

The concrete below the axis is doing nothing at all here — that is what cracked means, and it is why the steel stress is so much the larger number.

Thick Walls & Discs

Pressure through a thick wall, and the hole that doubles a spinning disc.

Where it is. 13th of 18 in the sidebar. It stands alone: nothing has to be set up before it.

WHAT IT IS FOR

Two constants fixed by the two pressures. The equilibrium equation it satisfies is the strongest test in the whole app.

$$\sigma_r=A-B/r^2 \quad \sigma_\theta=A+B/r^2 \quad \sigma_r(r_i)=-p_i, \quad \sigma_r(r_o)=-p_o$$

A pinhole at the centre doubles the peak stress. The limit is genuinely discontinuous, so it is offered as two cases.

$$\sigma_\theta=\rho\omega^2[\frac{3+\nu}{8}(r_o^2+r_i^2+r_i^2r_o^2/r^2)-\frac{(1+3\nu)}{8}r^2] \quad \sigma_\theta(\theta,ring)=\rho\omega^2r^2$$

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
r inner	mm	10.0 to 600 mm	100.0 mm
r outer	mm	10.0 to 600 mm	250.0 mm
read at, through the wall	– (a ratio)	0 to 1.00	0
ω	rad/s	10.0 to 3000 rad/s	300.0 rad/s
ρ	kg/m³	1000 to 20000 kg/m³	7850 kg/m³
ν	– (a ratio)	0 to 0.490	0.3000
p internal	MPa	0 to 300 MPa	40.00 MPa
p external	MPa	0 to 300 MPa	0 MPa
t at the axis	mm	2.00 to 200 mm	40.00 mm
design stress	MPa	10.0 to 800 MPa	150.0 MPa
junction, through the wall	– (a ratio)	0.0500 to 0.950	0.5000
radial interference	μm	1.00 to 500 μm	50.00 μm
E	GPa	50.0 to 400 GPa	200.0 GPa
ν	– (a ratio)	0 to 0.490	0.3000

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
r	mm	
σ hoop	MPa	Stress running around the circumference of a shell or disc.
σ radial	MPa	Stress acting along the radius.
peak hoop	MPa	
same disc, solid	MPa	
thin ring at the rim	MPa	Hoop stress in a thin spinning ring.
t at the rim	mm	Thickness a spinning disc needs at each radius to hold one stress everywhere.
A	MPa	Constant part of the radial stress distribution in a thick cylinder.
B	N	Coefficient of the inverse-square part of that distribution.
at the bore	MPa	
at the outside	MPa	
u radial, at the station	μm	Outward movement of a point in a thick cylinder.
r junction	mm	
p contact	MPa	Pressure generated at the interface of a shrink fit.
interference recovered	μm	Radial misfit built into a shrink fit before assembly.

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	r_inner	m	Inner radius of a thick cylinder, curved bar or disc.
IN	r_outer	m	Outer radius of a thick cylinder, curved bar or disc.
IN	p_int	Pa	Pressure acting on the inside surface.
IN	p_ext	Pa	Pressure acting on the outside surface.
IN	ρ_{rad}	m	Radius of the particular point being examined.
IN	E	Pa	Stiffness in simple tension: stress divided by strain while the response is still straight.
IN	ν	– (a ratio)	How much the material narrows sideways per unit of stretch along the pull.
IN	r_junction	m	Radius at which the two parts of a compound cylinder meet. Interface radius of a compound cylinder
IN	ρ_{mat}	kg/m^3	Mass of the material per unit volume.
IN	ω	rad/s	Rotation rate in radians per second.
IN	σ_{design}	Pa	Uniform stress a constant-strength profile is designed to hold everywhere. Uniform stress targeted by a constant-strength profile
OUT	lame_A	Pa	Constant part of the radial stress distribution in a thick cylinder.
OUT	lame_B	$\text{Pa} \cdot \text{m}^2$	Coefficient of the inverse-square part of that distribution.
OUT	σ_{r}	Pa	Stress acting along the radius.
OUT	σ_{hoop}	Pa	Stress running around the circumference of a shell or disc.
OUT	u_radial	m	Outward movement of a point in a thick cylinder.
OUT	p_contact	Pa	Pressure generated at the interface of a shrink fit.
OUT	interference	m	Radial misfit built into a shrink fit before assembly.
OUT	σ_{ring}	Pa	Hoop stress in a thin spinning ring.
OUT	t_profile	m	Thickness a spinning disc needs at each radius to hold one stress everywhere.

WHERE YOU HAVE TO CHOOSE

Which pressures act?

internal only | external only | both

External pressure reverses the sign of the hoop stress; the failure mode changes with it.

Thick Walls & Discs, the pressures panel: the p internal and p external sliders. Internal only, external only and both are the three ways of leaving one of them at zero, and the hoop stress reverses sign between the first two.

Solid disc or disc with a central hole?

solid | with a central hole

Not a matter of setting the bore to zero: the two answers differ by a factor of two exactly at that point.

Thick Walls & Discs, the spinning panel: the switch reading there is a central hole. The peak hoop stress doubles across it, and the decision card beneath says it is a different solution rather than a limit.

A WORKED TASK

A thick cylinder is pressurised from the inside. You want the hoop stress at the bore, where it is worst.

- 1. Set the inner and outer radii and the internal pressure.
- 2. Both stresses are plotted across the wall; drag the radius marker to read any point.

YOU TYPE		
r_i	100.0 mm	
r_o	200.0 mm	
p_i	60.00 MPa	
IT COMES BACK WITH		
hoop_bore	100.0 MPa	hoop stress at the bore
hoop_out	40.00 MPa	and at the outside face
radial_bore	-60.00 MPa	radial stress at the bore

The radial stress at the bore is exactly minus the pressure — it has to be, that face is being pushed on. Use it as your check that the radii went in the right way round.

Curved Beams

The neutral axis is not the centroid, and the error if you forget.

Where it is. 14th of 18 in the sidebar. It stands alone: nothing has to be set up before it.

WHAT IT IS FOR

The neutral axis of a sharply curved bar sits inside the centroid, and the gap between them is where the arithmetic dies.

$$R=A/(\int dA/r) \quad R_{\text{rect}}=h/(\ln(r_o/r_i)) \quad \sigma=M(R-r)/(A r (\bar{r}-R))$$

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
r inner	mm	5.00 to 500 mm	40.00 mm
r outer	mm	5.00 to 500 mm	80.00 mm
M	N·m	−20000 to 20000 N·m	1200 N·m
N, axial	kN	−200 to 200 kN	0 kN

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
A	mm ²	
rbar	mm	
R neutral	mm	Radius at which the bending stress in a curved bar passes through zero.
e offset	mm	Small gap between the centroid and the neutral axis of a curved bar.
σ inner	MPa	Stress in a curved bar carrying both bending and a force along its centre line.
σ outer	MPa	
straight-beam inner	MPa	
error if you used it	%	

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	r_inner	m	Inner radius of a thick cylinder, curved bar or disc.
IN	r_outer	m	Outer radius of a thick cylinder, curved bar or disc.
IN	h_depth	m	Overall depth of a section measured in the bending direction.
IN	A	m ²	Cross-sectional area actually carrying the load.
IN	M_x	N·m	Internal bending moment at the cut.
IN	N_axial	N	Internal force acting along the member axis at the cut being examined.
IN	ρ_rad	m	Radius of the particular point being examined.
IN	shape_curved	– (a ratio)	Which section the curved bar has. Two of them have exact closed forms; the rest are integrated as bands of linearly varying width. Which section the bar has. The rectangle and the round bar have closed forms of their own; everything else is integrated as radial bands.
IN	b_outer	m	Width of the section at its outer face, for a shape whose width is not the same at both. Zero at one face is a triangle; equal at both is the rectangle.
IN	b_flange	m	Width of a flange measured out from the web.
IN	d_flange	m	How deep the flange reaches, measured out along the radius.
OUT	R_neutral	m	Radius at which the bending stress in a curved bar passes through zero.
OUT	e_offset	m	Small gap between the centroid and the neutral axis of a curved bar.
OUT	σ_curved	Pa	Bending stress in a sharply curved bar.
OUT	σ_curved_axial	Pa	Stress in a curved bar carrying both bending and a force along its centre line.
OUT	R_neutral_builtup	m	Radius of the neutral axis of a curved section built up out of bands -- the same quantity as for a plain rectangle, for a section that has no closed form of its own.
OUT	e_builtup	m	How far inside its own centroid the neutral axis of a built-up curved section sits.

WHERE YOU HAVE TO CHOOSE

Which section is being bent?

rectangle | round | trapezoid | tee, flange inside | tee, flange outside | I section

The rectangular closed form takes two radii and nothing else, so it accepts any section a reader has and answers for a solid rectangle spanning those radii instead. On a bar that is a rectangle with a triangle on it, that is 23 per cent more area, a neutral radius 2.7 per cent too far out, and -- because the stress carries the centroid-to-neutral offset in its denominator -- a third low at the outer tip. Nothing in its two arguments can tell that the question was about a different bar, so the section has to be asked for rather than assumed.

Curved Beams, the section panel: a picker naming the six sections. The dimension sliders below it change with the choice -- a trapezoid gets a second width, a tee gets a flange width and depth -- and the area, neutral radius and offset readouts move with them.

A WORKED TASK

A hook has a rectangular section and a tight radius. The straight-beam formula would be badly wrong; you want how wrong, and the real stress.

- 1. Set the inner and outer radii of the curve and the section.
- 2. The neutral axis is drawn where it actually is — pulled towards the centre of curvature, not at the centroid.

YOU TYPE		
r_i	50.00 mm	inner radius of the curve
r_o	100.0 mm	outer radius
IT COMES BACK WITH		
R_n	72.13 mm	radius of the neutral axis
e	2.865 mm	how far the neutral axis sits inside the centroid

That offset is small and it is the whole difference. A straight-beam calculation puts the neutral axis at the centroid and gets the inner-fibre stress badly low.

Helical Springs

One ratio decides everything, including how much the correction costs.

Where it is. 15th of 18 in the sidebar. It stands alone: nothing has to be set up before it.

WHAT IT IS FOR

A spring is a bar in torsion wrapped into a helix. The correction factor is where tight coils bite.

$$\delta=8PD^3n/(Gd^4) \quad k=Gd^4/(8D^3n) \quad K_w=(4C-1)/(4C-4)+0.615/C, \quad C=D/d$$

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
P	N	1.00 to 5000 N	250.0 N
D, coil	mm	4.00 to 200 mm	30.00 mm
d, wire	mm	0.500 to 40.0 mm	4.000 mm
n, active coils	– (a ratio)	2.00 to 40.0	8.000
G	GPa	20.0 to 120 GPa	79.00 GPa
k ₂ / k ₁	– (a ratio)	0.100 to 5.00	1.000

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
C	– (a ratio)	Coil diameter divided by wire diameter; small values mean a tightly wound spring.
K, Wahl	– (a ratio)	Correction that raises the spring stress because the inner side of the coil is more sharply curved.
δ	mm	Distance a spring closes up under its load.
k	kN/m	Force needed per unit closure of a spring.
U	J	Work stored in a compressed or extended spring.
P recovered	N	
τ, corrected	MPa	Shear stress in the spring wire, including the correction for coil curvature.
τ, plain torsion	MPa	
correction adds	%	
k, in series	kN/m	
k, in parallel	kN/m	

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	P	N	Force applied along the member axis, or the single load in a standard case.
IN	D_coil	m	Mean diameter of a spring coil, measured to the wire centre.
IN	d_wire	m	Diameter of the wire a spring is wound from.
IN	n_coils	– (a ratio)	Number of spring coils that actually deflect. Number of active coils
IN	G	Pa	Stiffness in shear: shear stress divided by the angle change it produces.
OUT	δ_spring	m	Distance a spring closes up under its load.
OUT	k_spring	N/m	Force needed per unit closure of a spring.
OUT	C_index	– (a ratio)	Coil diameter divided by wire diameter; small values mean a tightly wound spring.
OUT	K_wahl	– (a ratio)	Correction that raises the spring stress because the inner side of the coil is more sharply curved.
OUT	τ_spring	Pa	Shear stress in the spring wire, including the correction for coil curvature.
OUT	U_spring	J	Work stored in a compressed or extended spring.

WHERE YOU HAVE TO CHOOSE

Nothing on this screen has two defensible answers, so there is nothing to choose.

A WORKED TASK

A helical compression spring carries a load. You want how far it squashes, its rate, and the real shear stress in the wire.

1. Set the load, the coil and wire diameters, the number of active coils and the shear modulus.
2. The spring index is shown next to the diameters it comes from, because everything else follows from it.
3. Both the plain and the corrected stress are printed, with the difference between them.

YOU TYPE		
P	250.0 N	
D	30.00 mm	coil diameter
d	4.000 mm	wire diameter
n	8.000	active coils
G	79.00 GPa	
IT COMES BACK WITH		
C	7.500	spring index, coil over wire
K	1.197	correction for the wire being curved, not straight
δ	21.36 mm	how far it squashes
k	11.70 kN/m	spring rate
τ	357.3 MPa	shear stress in the wire, corrected
τ_plain	298.4 MPa	and what the straight-bar formula would have said

The correction is not a rounding error. Wind the spring more tightly — drop the coil diameter towards the wire diameter — and it grows fast.

Plastic Bending

Past first yield, and the stress the bar keeps afterwards.

Where it is. 16th of 18 in the sidebar. It reads the cross-section itself — its outline, area, centroid, second moments and extreme fibres from the Section Builder screen, so set that up first if you have not.

WHAT IT IS FOR

The fully plastic section splits at the equal-area axis, not the centroid.

$$Z = \sum |A_i \bar{y}_i| \quad f = Z/S \quad M_p = \sigma_Y Z$$

Unloading from a partly plastic state is elastic, so the difference locks in as residual stress.

$$M_Y = \sigma_Y S \quad M_p = \sigma_Y Z \quad \sigma_{res} = \sigma_{elastoplastic} - \sigma_{unload}$$

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
σ_Y	MPa	50.0 to 1200 MPa	250.0 MPa
M applied	kN·m	0 to 500 kN·m	45.00 kN·m
b	mm	5.00 to 500 mm	60.00 mm
h	mm	5.00 to 800 mm	120.0 mm

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
y_pna, from the centroid	mm	Position of the axis that divides the section into two equal areas.
y_pna, from the origin	mm	
y_c, from the origin	mm	
Z plastic	mm ³	Plastic section modulus; the first moment of the two halves about the equal-area axis.
S elastic	mm ³	
f, shape factor	— (a ratio)	Ratio of plastic to elastic section modulus; how much reserve the shape has past first yield.
M Y, first yield	kN·m	
M p, fully plastic	kN·m	Moment at which the whole section has yielded.
M applied is through	%	
f, rectangle	— (a ratio)	
M Y, rectangle	kN·m	
M p, rectangle	kN·m	
y core	mm	Half-depth of the region that is still elastic in a partly yielded section.
M recovered from the core	kN·m	
residual at the surface	MPa	Stress left locked in after loading past yield and unloading.
residual at the core edge	MPa	
residual couple, as a share of M	%	
net residual force	kN	

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	parts	– (a ratio)	Ordered list of primitive shapes that together make up the section.
IN	σ_Y	Pa	Stress at which the material stops behaving elastically.
IN	S_{mod}	m^3	Section modulus; second moment divided by the distance to the extreme fibre.
IN	$Z_{plastic}$	m^3	Plastic section modulus; the first moment of the two halves about the equal-area axis.
IN	M_x	$N \cdot m$	Internal bending moment at the cut.
IN	$c_{extreme}$	m	Distance from the neutral axis to the farthest fibre.
OUT	y_{pna}	m	Position of the axis that divides the section into two equal areas.
OUT	f_{shape}	– (a ratio)	Ratio of plastic to elastic section modulus; how much reserve the shape has past first yield.
OUT	M_p	$N \cdot m$	Moment at which the whole section has yielded.
OUT	M_Y	$N \cdot m$	Moment at which the extreme fibre first reaches yield.
OUT	$y_{elastic_core}$	m	Half-depth of the region that is still elastic in a partly yielded section.
OUT	$\sigma_{residual}$	Pa	Stress left locked in after loading past yield and unloading.

WHERE YOU HAVE TO CHOOSE

Nothing on this screen has two defensible answers, so there is nothing to choose.

A WORKED TASK

A rectangular bar is bent past first yield. You want the moment at first yield and the moment when the whole section has gone.

1. Set the rectangle and the yield stress.
2. The stress block is drawn as it spreads inward from both faces.

YOU TYPE		
b	50.00 mm	
h	100.0 mm	
σ_Y	250.0 MPa	
IT COMES BACK WITH		
Z_p	125000 mm ³	plastic section modulus
M_p	31.25 kN·m	fully plastic moment
M_Y	20.83 kN·m	moment at first yield
f	1.500	how much is left after first yield

For a rectangle the shape factor is one and a half: the section carries half as much again after the outer fibre has yielded. Take the load off and the bar keeps a residual stress, which the screen also draws.

Riveted & Welded Joints

Three ways to fail, and the one that decides.

Where it is. 17th of 18 in the sidebar. It stands alone: nothing has to be set up before it.

WHAT IT IS FOR

Three ways to fail, and the joint is only as strong as the weakest. Naming the winner is the whole point.

$$P_s = n(\pi d^2/4)\tau$$
$$P_b = n d t \sigma_b$$
$$P_t = (p-d) t \sigma_t$$
$$\eta = \min(P_s, P_b, P_t) / (p t \sigma_t)$$

A fillet weld fails across its throat. Eccentric loading adds a torsional component that must be summed as a vector.

$$t_{throat} = 0.707 s$$
$$P = 0.707 s L_w$$
$$\tau_{allow}$$
$$\tau_{res} = [\tau_{direct} + \tau_{torsional}]$$

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
n rivets per pitch	– (a ratio)	1.00 to 12.0	4.000
d rivet	mm	4.00 to 50.0 mm	20.00 mm
t plate	mm	2.00 to 40.0 mm	10.00 mm
pitch	mm	10.0 to 300 mm	60.00 mm
shear planes	– (a ratio)	1.00 to 2.00	1.000
τ allow, rivet	MPa	10.0 to 400 MPa	90.00 MPa
σ bearing	MPa	20.0 to 800 MPa	200.0 MPa
σ tension, plate	MPa	20.0 to 600 MPa	150.0 MPa
leg size	mm	2.00 to 20.0 mm	6.000 mm
run length	mm	20.0 to 600 mm	150.0 mm
τ allow	MPa	10.0 to 400 MPa	90.00 MPa
P	kN	0 to 300 kN	30.00 kN
e, eccentricity	mm	0 to 500 mm	80.00 mm

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
P shear	kN	Strength of one pitch length if the fasteners fail by being cut through.
P bearing	kN	Strength of one pitch length if the plate crushes against the fasteners.
P tearing	kN	Strength of one pitch length if the plate tears across the line of holes.
P joint	kN	Strength of one pitch length, being the smallest of the three failure modes. Multiply by the number of pitches for the whole seam.
P solid plate, per pitch	kN	
efficiency	%	Strength of one pitch length divided by the strength of the same length of unbroken plate. All four quantities are per pitch, so the ratio is the same for the whole seam.
Unwin's starting diameter	mm	
throat	mm	Shortest distance across a fillet weld, where it actually fails.
P weld	kN	Force a weld run may carry across its throat.
J weld	mm ⁴	Polar second moment of a weld group about its own centroid.
τ direct only	MPa	
τ combined	MPa	Peak shear stress across the weld throat.

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	d_rivet	m	Diameter of the rivet or bolt.
IN	t_plate	m	Thickness of the plate being joined.
IN	pitch	m	Centre-to-centre spacing of fasteners along a row. Centre-to-centre spacing along a row. Every strength on this screen is the strength of one such length of seam.
IN	n_rivets	– (a ratio)	Number of fasteners in one pitch length, counting every row. A double-riveted joint has two, whatever the length of the seam. Per pitch length, counting every row -- the same basis the tearing strength and the efficiency denominator are on.
IN	τ_allow	Pa	Highest shear stress permitted in service.
IN	σ_bear	Pa	Highest bearing pressure permitted between a fastener and the plate.
IN	σ_tens	Pa	Highest tensile stress permitted in the plate.
IN	shear_planes	– (a ratio)	Number of surfaces across which a fastener is cut.
IN	s_leg	m	Leg length of a fillet weld, measured along the joined faces.
IN	L_weld	m	Effective length of weld run resisting the load.
IN	P	N	Force applied along the member axis, or the single load in a standard case.
IN	e_ecc	m	Distance from the centroid to the line of action of the load.
IN	weld_group	– (a ratio)	Layout of the individual weld runs making up a group.
OUT	P_shear	N	Strength of one pitch length if the fasteners fail by being cut through.
OUT	P_bearing	N	Strength of one pitch length if the plate crushes against the fasteners.
OUT	P_tearing	N	Strength of one pitch length if the plate tears across the line of holes.
OUT	P_joint	N	Strength of one pitch length, being the smallest of the three failure modes. Multiply by the number of pitches for the whole seam.
OUT	η_joint	– (a ratio)	Strength of one pitch length divided by the strength of the same length of unbroken plate. All four quantities are per pitch, so the ratio is the same for the whole seam.
OUT	failure_mode	– (a ratio)	Which of the competing failure modes limits the joint.
OUT	t_throat	m	Shortest distance across a fillet weld, where it actually fails.
OUT	P_weld	N	Force a weld run may carry across its throat.
OUT	J_weld	m ⁴	Polar second moment of a weld group about its own centroid.
OUT	τ_weld	Pa	Peak shear stress across the weld throat.

WHERE YOU HAVE TO CHOOSE

Which failure mode governs?

rivet shear | plate bearing | plate tearing

The result must be a stated conclusion, not three numbers side by side. Reading the wrong one over-predicts strength.

Riveted & Welded Joints, the panel headed the three ways it can fail: the shear, bearing and tearing capacities on three lines, the joint capacity beneath them, and a decision card naming the governing mode and how much stronger the next one is.

A WORKED TASK

A riveted lap joint can fail three ways. You want which one governs, and how efficient the joint is.

1. Set the rivet diameter, the plate thickness, the pitch and the three allowable stresses.
2. All three strengths are printed side by side and the smallest is named as the governing mode.

YOU TYPE		
d	20.00 mm	rivet diameter
t	12.00 mm	plate thickness
p	80.00 mm	pitch
τ_a	80.00 MPa	allowable shear
σ_b	160.0 MPa	allowable bearing
σ_t	120.0 MPa	allowable tension

IT COMES BACK WITH		
P _s	25.13 kN	shearing the rivet
P _b	38.40 kN	crushing the plate against the rivet
P _t	86.40 kN	tearing the plate between holes
P	25.13 kN	what the joint actually carries

The smallest of the three is the answer and the screen says which one it was. Here it is the rivet, and tearing is more than three times stronger. Now drag the pitch down towards thirty-five millimetres: tearing falls below the rivet and takes over as the governing mode, without a single allowable stress changing. That is the point of showing all three.

Axial & Thermal

Bars, temperature, misfit and the gap that changes everything.

Where it is. 18th of 18 in the sidebar. It stands alone: nothing has to be set up before it.

WHAT IT IS FOR

One force over one area. Every later screen reduces to this at the last step.

$$\sigma=P/A \quad \tau_{avg}=V/A \quad \epsilon=\delta/L \quad n=\sigma_{fail}/\sigma_{allow}$$

Any two of E, G, K and nu fix the other two. Six inversion paths that must all agree.

$$G=E/(2(1+\nu)) \quad K=E/(3(1-2\nu)) \quad e_{\nu}=((1-2\nu)/E)(\sigma_x+\sigma_y+\sigma_z)$$

A bar pulled along one axis still carries shear on every other plane. The bridge to the transformation screens.

$$\sigma_{\theta}=\sigma_x \cos^2\theta \quad \tau_{\theta}=- (1/2)\sigma_x \sin 2\theta$$

The same weight applied three ways gives three different stresses. Which way is a choice the user must make.

$$U=P^2L/(2AE) \quad n_{imp}=1+\sqrt{(1+2h/\delta_{st})}$$

Four shapes of the same integral. Each degenerates into the prismatic case, and the degeneracy is where the arithmetic breaks.

$$\delta=PL/(AE) \quad \delta_{cone}=4PL/(\pi E d_1 d_2) \quad \delta_{wedge}=PL \ln (b_2/b_1)/(E t (b_2-b_1)) \quad \delta_{self}=\gamma L^2/(2E)$$

Equilibrium alone is not enough; compatibility supplies the missing equation.

$$\delta_1=\delta_2 \text{ and } P_1+P_2=P \implies \sigma_1/E_1=\sigma_2/E_2, \quad \sigma_1A_1+\sigma_2A_2=P$$

A bar that cannot expand carries stress with no load on it at all.

$$\delta_T=\alpha \Delta T L \quad \sigma_{restrained}=-E \alpha \Delta T \quad \sigma_{gap}=-E(\alpha \Delta T L-g)/L$$

A geometric discontinuity multiplies the nominal stress. Two shapes have exact closed forms; the chart families do not, and are not guessed at.

$$\sigma_{\theta}(a,\theta)=\sigma(1-2 \cos 2\theta) \quad K_t=3 \quad K_t=1+2a/b=1+2\sqrt{(a/\rho)}$$

In a cylinder the circumferential stress is exactly twice the longitudinal one, always.

$$\sigma_{hoop}=pr/t \quad \sigma_{long}=pr/(2t) \quad \sigma_{sphere}=pr/(2t)$$

WHAT YOU SET

ON SCREEN	UNIT	RANGE	OPENS AT
P	kN	−500 to 500 kN	60.00 kN
A	mm ²	10.0 to 20000 mm ²	500.0 mm ²
L	m	0.100 to 20.0 m	2.000 m
d1	mm	5.00 to 100 mm	40.00 mm
d2	mm	5.00 to 100 mm	20.00 mm
b1	mm	5.00 to 100 mm	60.00 mm
b2	mm	5.00 to 100 mm	30.00 mm
t	mm	0.500 to 50.0 mm	10.00 mm
γ	kN/m ³	1.00 to 100 kN/m ³	77.00 kN/m ³
V	kN	0 to 500 kN	5.000 kN
D	mm	3.00 to 50.0 mm	20.00 mm
t	mm	1.00 to 30.0 mm	10.00 mm
E	GPa	1.00 to 600 GPa	200.0 GPa
ν	— (a ratio)	−0.990 to 0.490	0.3000
σ_Y	MPa	20.0 to 1200 MPa	250.0 MPa
A ₁	mm ²	10.0 to 20000 mm ²	600.0 mm ²
E ₁	GPa	5.00 to 400 GPa	200.0 GPa
A ₂	mm ²	0 to 20000 mm ²	400.0 mm ²
E ₂	GPa	5.00 to 400 GPa	70.00 GPa
ΔT	K	−300 to 300 K	60.00 K
α	1e−6/K	0.500 to 30.0 1e−6/K	12.00 1e−6/K
gap	mm	0 to 5.00 mm	0 mm
drop height	mm	0 to 500 mm	0 mm
p	MPa	0 to 30.0 MPa	1.500 MPa
r	mm	20.0 to 3000 mm	500.0 mm
t	mm	0.500 to 50.0 mm	8.000 mm
L, vessel	mm	100 to 20000 mm	2000 mm
a, across the pull	mm	0.500 to 100 mm	10.00 mm
b, along the pull	mm	0.500 to 100 mm	10.00 mm
θ round the edge	deg	0 to 180 deg	—
P	kN	−500 to 500 kN	—
A	mm ²	10.0 to 20000 mm ²	—
L	m	0.100 to 20.0 m	—
E	GPa	5.00 to 400 GPa	—

A dash in the last two columns is a control that belongs to a row you add yourself — a load, a part, a segment — or one whose limits follow another input, so there is no single value to print. The screen still shows both as soon as the row exists.

WHAT IT TELLS YOU

READS	UNIT	WHAT IT IS
closes at ΔT	K	Whether the bar has taken up its clearance and begun to carry stress.
laid on gently	MPa	
released from rest	MPa	
dropped	MPa	
impact factor	– (a ratio)	How many times larger the effect is than the same load applied gently.
σ	MPa	Force per unit area acting square to the cut.
δ	mm	
ϵ	– (a ratio)	Change in length divided by the original length.
k	N/m	Force needed per unit stretch.
n	– (a ratio)	How many times the applied action could be increased before failure.
U	J	Work stored in a bar stretched along its axis.
u	J/m ³	Work stored per unit volume.
τ_{avg}	MPa	Shear force spread evenly over the area it crosses.
σ_b	MPa	Pressure between two parts pressed together.
G	Pa	Stiffness in shear: shear stress divided by the angle change it produces.
K	Pa	Resistance to uniform squeezing: pressure divided by fractional volume change.
λ	Pa	First of the two elastic parameters that state the three-dimensional relations most compactly.
ν recovered from E and G	– (a ratio)	
E recovered from K and G	Pa	
ϵ_{th}	– (a ratio)	Strain the material would take up freely on a temperature change.
σ , fully restrained	MPa	Stress caused purely by preventing free thermal movement.
σ , with the gap	MPa	Thermal stress once the bar has closed a clearance and met its restraint.
k, both together	N/m	Force needed per unit stretch.
sphere	MPa	Membrane stress in a spherical shell, the same in every direction.
$\Delta V/V$	– (a ratio)	Fractional change in the volume a shell encloses.
Δr	mm	Change in radius of a pressurised shell.
ΔL	mm	Change in length of a pressurised shell.
σ nominal	MPa	
σ at the edge, at θ	MPa	
K t, round hole	– (a ratio)	Concentration factor for a small round hole in a wide plate.
K t, least round the edge	– (a ratio)	The smallest hoop stress on a hole's edge, as a multiple of the applied tension. It is negative: the edge is in compression along the load even though the plate is pulled.
K t, this ellipse	– (a ratio)	Concentration factor for an elliptical hole, sharper as the ellipse flattens.
ρ , tip radius	mm	Tip radius of an elliptical hole: how sharply the hole turns at the end of the axis across the load. A crack is this radius going to zero.
K t, from that tip radius	– (a ratio)	
σ_{max}	MPa	Largest normal stress present.
free expansion	mm	The bar carries stress only once the first exceeds the second.
the gap	mm	
P ₁	N	The stiffer member takes the larger share. Each is computed on its own, not by subtracting one from the load.
P ₂	N	

σ_1	MPa	Equal moduli would give one stress in both; different moduli cannot.
σ_2	MPa	
δ_1	mm	
δ_2	mm	
hoop	MPa	Exactly twice, always, for a cylinder -- which is why a pressure vessel splits along its length and not around it.
longitudinal	MPa	

EVERY QUANTITY ON THIS SCREEN, DEFINED

	SYMBOL	UNIT	WHAT IT MEANS
IN	P	N	Force applied along the member axis, or the single load in a standard case. Internal axial force at the cut, not the applied load
IN	A	m ²	Cross-sectional area actually carrying the load. Area actually carrying the force
IN	V	N	Internal shear force at the cut.
IN	A _b	m ²	Projected contact area over which one part presses on another. Projected contact area for bearing
IN	L	m	Length of the member along its axis.
IN	δ	m	Change in length of the member.
IN	σ _{fail}	Pa	Stress at which the part is taken to have failed.
IN	σ _{allow}	Pa	Highest normal stress permitted in service.
IN	E	Pa	Stiffness in simple tension: stress divided by strain while the response is still straight. Slope of the straight part of the tensile response
IN	ν	– (a ratio)	How much the material narrows sideways per unit of stretch along the pull. Sideways contraction per unit stretch
IN	G	Pa	Stiffness in shear: shear stress divided by the angle change it produces.
IN	K _{bulk}	Pa	Resistance to uniform squeezing: pressure divided by fractional volume change.
IN	σ _x	Pa	Normal stress on the face whose outward normal points along the first axis.
IN	σ _y	Pa	Normal stress on the face whose normal points along the second axis.
IN	σ _z	Pa	Normal stress on the face whose normal points along the third axis.
IN	θ	rad	Angle through which an element or a cutting plane has been rotated. Angle from the bar axis to the cut normal
IN	h _{drop}	m	Height a weight falls through before it strikes. Free-fall height before contact; zero means suddenly applied
IN	δ _{st}	m	Deflection the same load would cause if it were lowered gently into place. Deflection the same load would cause if applied gradually
IN	d1	m	Diameter at one end of a round bar that tapers. Diameter at one end of a tapered round bar
IN	d2	m	Diameter at the other end of a round bar that tapers.
IN	b1	m	Width at one end of a bar that tapers in width. Width at one end of a tapered flat bar
IN	b2	m	Width at the other end of a bar that tapers in width.
IN	t _{thk}	m	Constant thickness of a flat bar that tapers only in width. Constant thickness of the flat bar
IN	γ _{wt}	N/m ³	Weight of the material per unit volume.
IN	segments	– (a ratio)	List of prismatic pieces making up a stepped member. List of prismatic segments, each with its own P, L, A and E
IN	A1	m ²	Area of the first element in a two-element assembly.
IN	A2	m ²	Area of the second element in a two-element assembly.
IN	E1 _{mat}	Pa	Stiffness of the first material in a two-material assembly.
IN	E2 _{mat}	Pa	Stiffness of the second material.
IN	α _{th}	1/K	Fractional change in length per degree of temperature change.
IN	dT	K	Temperature change, taken as positive for heating. Temperature change, signed
IN	gap	m	Free clearance a bar can take up before it meets its restraint. Free clearance before the bar meets its restraint
IN	σ _{nom}	Pa	Stress worked out ignoring the local geometric disturbance.
IN	a _{ell}	m	Half-width of an elliptical hole measured across the load direction. Semi-axis of the hole across the load
IN	b _{ell}	m	Half-width of an elliptical hole measured along the load direction. Semi-axis of the hole along the load
IN	ρ _{tip}	m	Tip radius of an elliptical hole: how sharply the hole turns at the end of the axis across the load. A crack is this radius going to zero. Tip radius, the other way to state the same ellipse
IN	p _{int}	Pa	Pressure acting on the inside surface.
IN	r _{mid}	m	Mid-wall radius of a thin shell.

IN	t _{wall}	m	Wall thickness of a tube or shell.
IN	η_{joint}	– (a ratio)	Strength of one pitch length divided by the strength of the same length of unbroken plate. All four quantities are per pitch, so the ratio is the same for the whole seam. Efficiency of a longitudinal seam, if any
OUT	σ	Pa	Force per unit area acting square to the cut.
OUT	τ_{avg}	Pa	Shear force spread evenly over the area it crosses.
OUT	σ_{b}	Pa	Pressure between two parts pressed together.
OUT	ϵ	– (a ratio)	Change in length divided by the original length.
OUT	n _{safety}	– (a ratio)	How many times the applied action could be increased before failure.
OUT	A _{req}	m ²	Smallest area that keeps the stress at or below what is allowed.
OUT	e _{vol}	– (a ratio)	Fractional change in volume.
OUT	λ_{lame}	Pa	First of the two elastic parameters that state the three-dimensional relations most compactly.
OUT	σ_{θ}	Pa	Normal stress on a plane cut at an angle across a bar in simple tension.
OUT	τ_{θ}	Pa	Shear stress on a plane cut at an angle across a bar in simple tension.
OUT	U _{axial}	J	Work stored in a bar stretched along its axis.
OUT	u _{density}	J/m ³	Work stored per unit volume.
OUT	n _{impact}	– (a ratio)	How many times larger the effect is than the same load applied gently.
OUT	σ_{impact}	Pa	Peak stress reached under a suddenly applied or dropped load.
OUT	δ_{impact}	m	Peak deflection reached under a suddenly applied or dropped load.
OUT	δ_{cone}	m	Stretch of a bar whose round section tapers along its length.
OUT	δ_{wedge}	m	Stretch of a flat bar whose width tapers along its length.
OUT	δ_{self}	m	Stretch a hanging bar suffers under nothing but its own weight.
OUT	k _{axial}	N/m	Force needed per unit stretch.
OUT	δ_{stepped}	m	Total stretch of a bar made of several prismatic pieces.
OUT	P1	N	Share of the load carried by the first element.
OUT	P2	N	Share carried by the second element.
OUT	$\sigma_{1\text{m}}$	Pa	Stress in the first material of a composite member.
OUT	$\sigma_{2\text{m}}$	Pa	Stress in the second material of a composite member.
OUT	ϵ_{th}	– (a ratio)	Strain the material would take up freely on a temperature change.
OUT	δ_{th}	m	Free length change a temperature change would produce if nothing resisted it.
OUT	σ_{th}	Pa	Stress caused purely by preventing free thermal movement.
OUT	$\sigma_{\text{th_gap}}$	Pa	Thermal stress once the bar has closed a clearance and met its restraint.
OUT	gap _{closed}	– (a ratio)	Whether the bar has taken up its clearance and begun to carry stress.
OUT	K _{t_hole}	– (a ratio)	Concentration factor for a small round hole in a wide plate.
OUT	K _{t_min}	– (a ratio)	The smallest hoop stress on a hole's edge, as a multiple of the applied tension. It is negative: the edge is in compression along the load even though the plate is pulled.
OUT	K _{t_ellipse}	– (a ratio)	Concentration factor for an elliptical hole, sharper as the ellipse flattens.
OUT	K _t	– (a ratio)	How many times the local stress exceeds the nominal stress at a geometric disturbance.
OUT	σ_{max}	Pa	Largest normal stress present.
OUT	σ_{hoop}	Pa	Stress running around the circumference of a shell or disc.
OUT	σ_{long}	Pa	Stress running along the axis of a cylindrical shell.
OUT	σ_{sphere}	Pa	Membrane stress in a spherical shell, the same in every direction.
OUT	dr _{change}	m	Change in radius of a pressurised shell.
OUT	dL _{change}	m	Change in length of a pressurised shell.

OUT	dV_ratio	– (a ratio)	Fractional change in the volume a shell encloses.
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WHERE YOU HAVE TO CHOOSE

Is the fastener in single or double shear?

single: $\tau = V/A$ | double: $\tau = V/(2A)$

A factor of two, and the commonest slip in the whole subject. Never pick for the user.

Axial & Thermal, the fastener panel: a two-segment control reading single / double. Switching it halves the shear stress readout while the bearing stress beside it stays where it was.

How is the load applied?

gradually: factor 1 | suddenly: factor 2 | dropped from height h: factor $1 + \sqrt{1 + 2h/\delta_{st}}$

Three separate sections in every textbook and a factor of two or more between them.

Axial & Thermal, the readout panel headed the same weight, three ways: laid on gently, released from rest and dropped are three readouts one under the other, with the impact factor beneath them and the drop-height slider feeding the third.

How is the bar restrained?

free to expand | fully restrained | restrained after closing a gap g

The stress jumps from zero to the full restrained value; the user must see which case applies.

Axial & Thermal, the temperature panel: the gap slider. At zero the bar is restrained from the first degree; above the free expansion nothing restrains it; between the two the gap closes part way. The decision card headed the gap decides everything names the case that holds, and the stress readout jumps with it.

Cylinder or sphere?

cylinder | sphere

The sphere carries half the cylinder's peak stress for the same pressure and wall.

Axial & Thermal, the vessel panel: hoop and longitudinal as a pair for the cylinder, and the sphere's stress on the line beneath, all from the same pressure, radius and wall.

A WORKED TASK

A bar is pulled along its length. You want the stress in it and how much it stretches.

1. Set the force, the area, the length and the modulus.
2. The same panel handles a temperature change and a misfit; leave them at zero for a plain pull.

YOU TYPE		
P	80.00 kN	
A	600.0 mm ²	
L	2.500 m	
E	200.0 GPa	
IT COMES BACK WITH		
σ	133.3 MPa	direct stress
δ	1.667 mm	how much longer it gets
k	48000 kN/m	stiffness of the bar

Now hold both ends and warm the bar instead. Nothing moves, and the whole of that stress appears anyway — which is what the thermal panel is for.

What is not in this manual

Screen photographs. This build's manual carries none. Every screen is described here in full — what it asks for, what it gives back, and a worked task — but the pictures are not in yet.

They cannot be faked and they cannot be automated around. Store and manual screenshots have to be driven through the real interface — a release build, clicked and typed as a person would — and never by presetting the interface from outside it. That rule is not a preference: an environment-variable hook left in for screenshot automation is what earned this developer account a Guideline 5.6 finding, and the binary now carries nothing that could be used that way.

One relation with no screen in this release. The specification behind this app holds more relations than this release puts a screen in front of, and the theory manual writes up every one of them. This one is fully documented there and reachable nowhere here, so counting screens and counting relations gives two different numbers — on purpose:

- **Stress-Strain Curve Reduction.** Reducing a measured stress-strain curve means handing the app a whole curve, and reading a file of your own test data is the one input route this release does not have. The relation itself is implemented and verified to the same standard as everything else here, and it is written up in full in the theory manual; what is missing is the way to get your own measurements in front of it.

18 screens · 52 of the 53 relations in the specification · 158 controls and 303 readouts described · 18 worked tasks, every answer computed by the same code the application runs · 0 screen photographs. Generated from the same specification the application carries. The theory manual documents all 53, including the one named above that this release has no screen for.

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